EC 571: Digital VLSI Circuit Design

Inverter VTC, MOS Capacitance

Ajay Joshi

Acknowledgement - Some slides adapted from:

- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

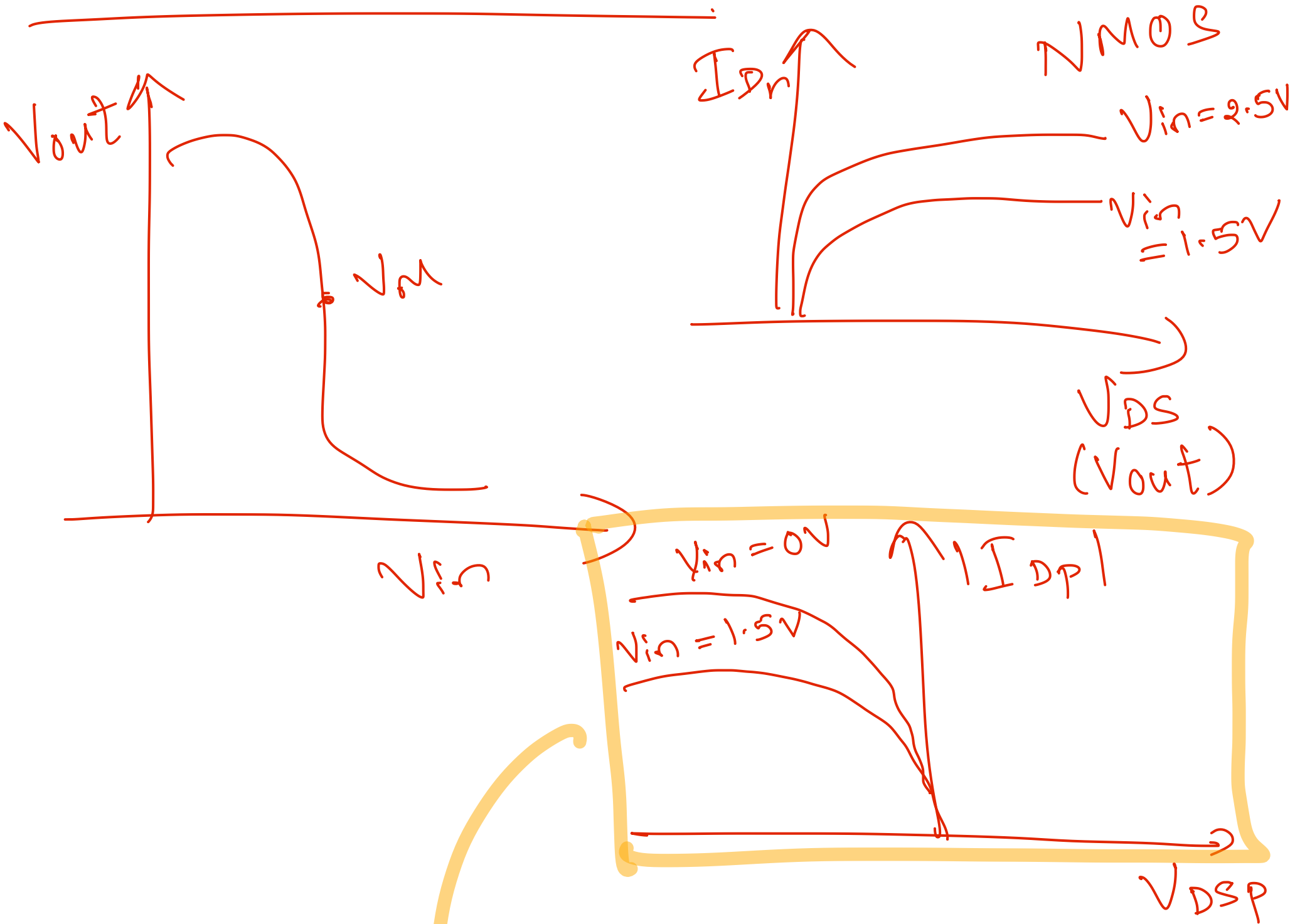
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Last lecture

- Last lecture
  - MOS Transistor model
- Today's lecture
  - Inverter VTC: Using MOS model
  - MOS Capacitance
- Reading
  - Chapter 3

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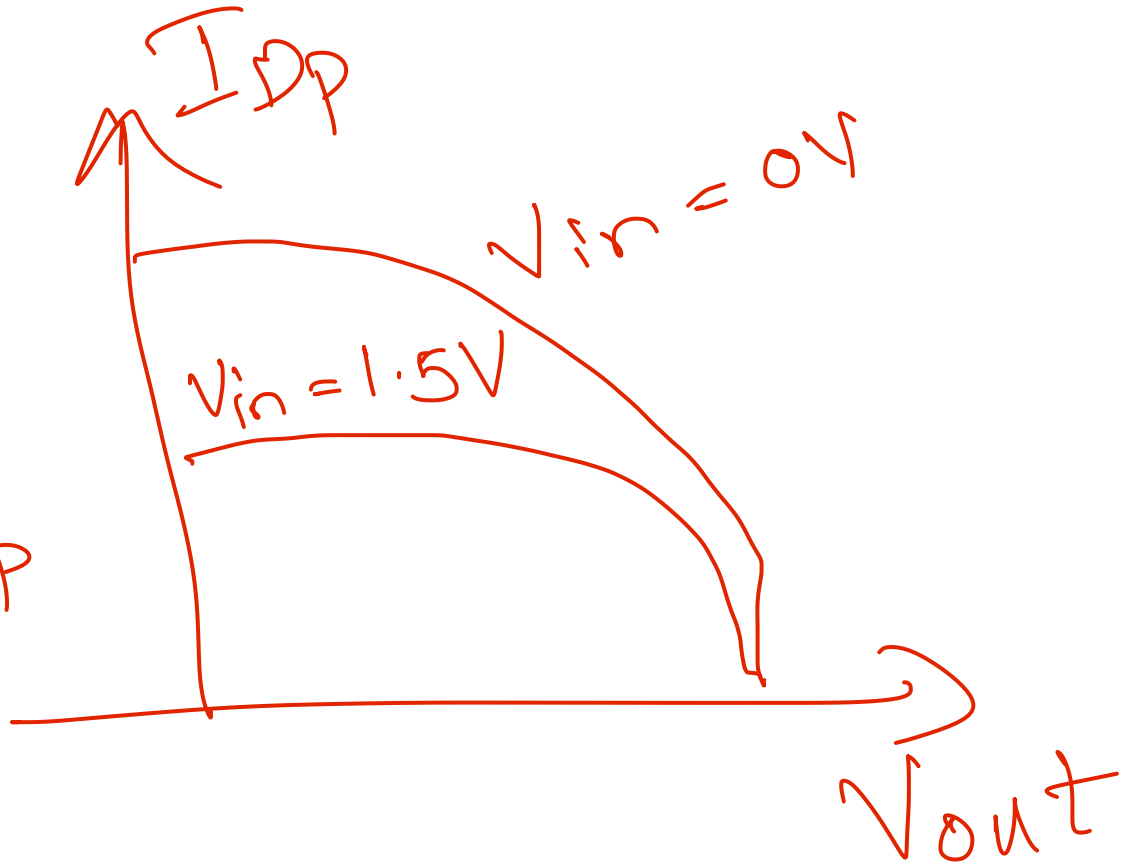
Inverter VTC

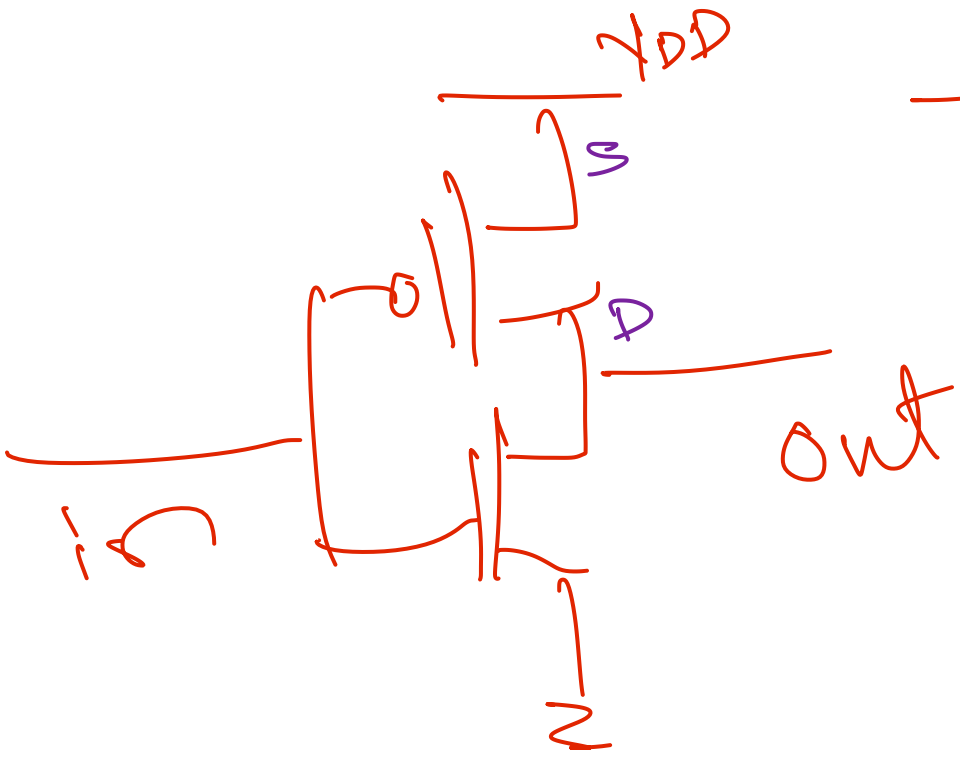


$V_{SDp} = -V_{DSp}$

$V_{out} = V_{DD} - V_{SDp}$

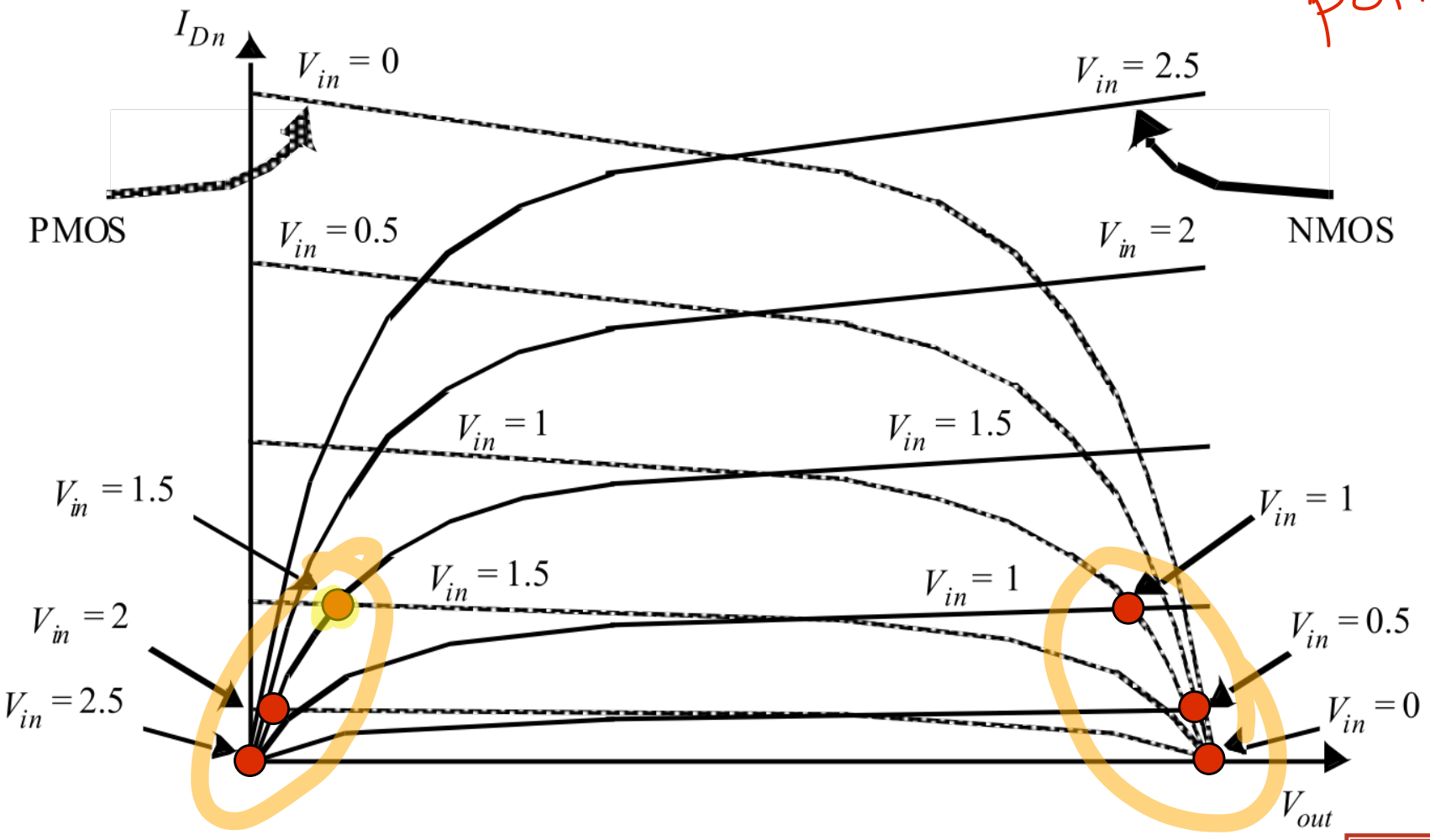
$V_{out} = V_{DD} + V_{DSp}$



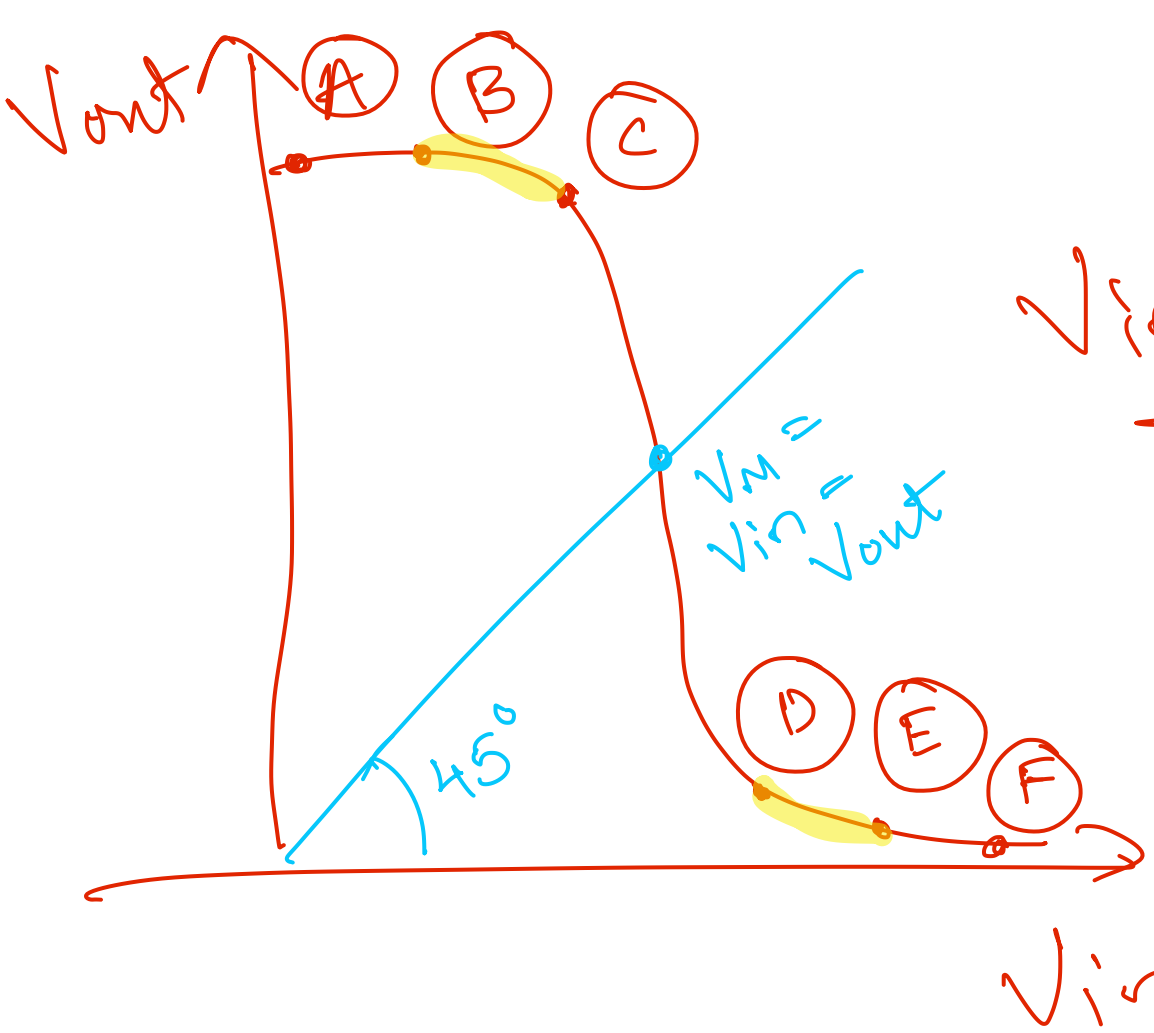


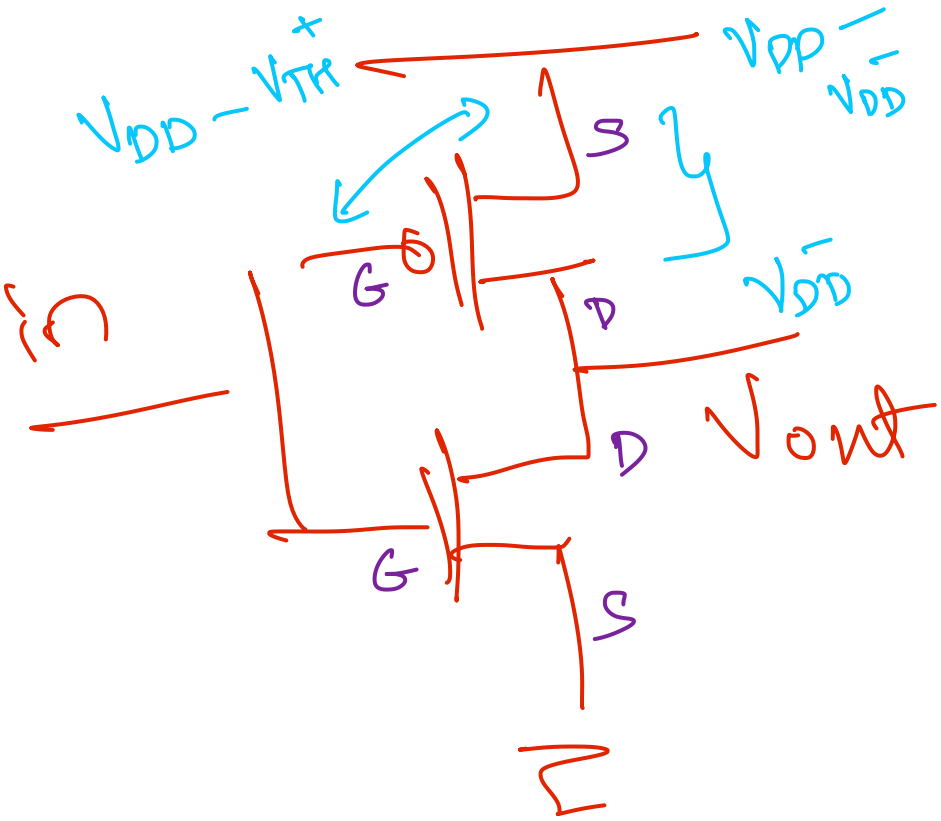
CMOS Inverter Load Characteristics

We are interested in the intersection points.



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$V_{DD} = 2.5V$

$V_{TH} = 0.5V$

A ⟷ B

NMOS $V_{GS_n} \rightarrow 0 < V_{TH}$

Cutoff

PMOS $V_{SDP} \rightarrow 0$ $|V_{GS_p}| = 2.5V$

$|V_{GS_p} - V_{TH_p}| \; V_{DS_p}$

$2.5 - 0.5 > 0$

Linear

(B) ⟷ (C)

V_{GS} is slightly larger than V_{TH}

NMOS $V_{GS} > V_{TH}$

$V_{DS} \rightarrow V_{DD}$

$V_{DS} > V_{GS} - V_{TH}$

Saturation

PMOS

$V_{GS_p} \rightarrow -2.0V$

$|V_{GS_p}| = 2V$

$V_{DS_p} = -0.2V \text{ to } -0.3V$

Linear

C ⟷ D

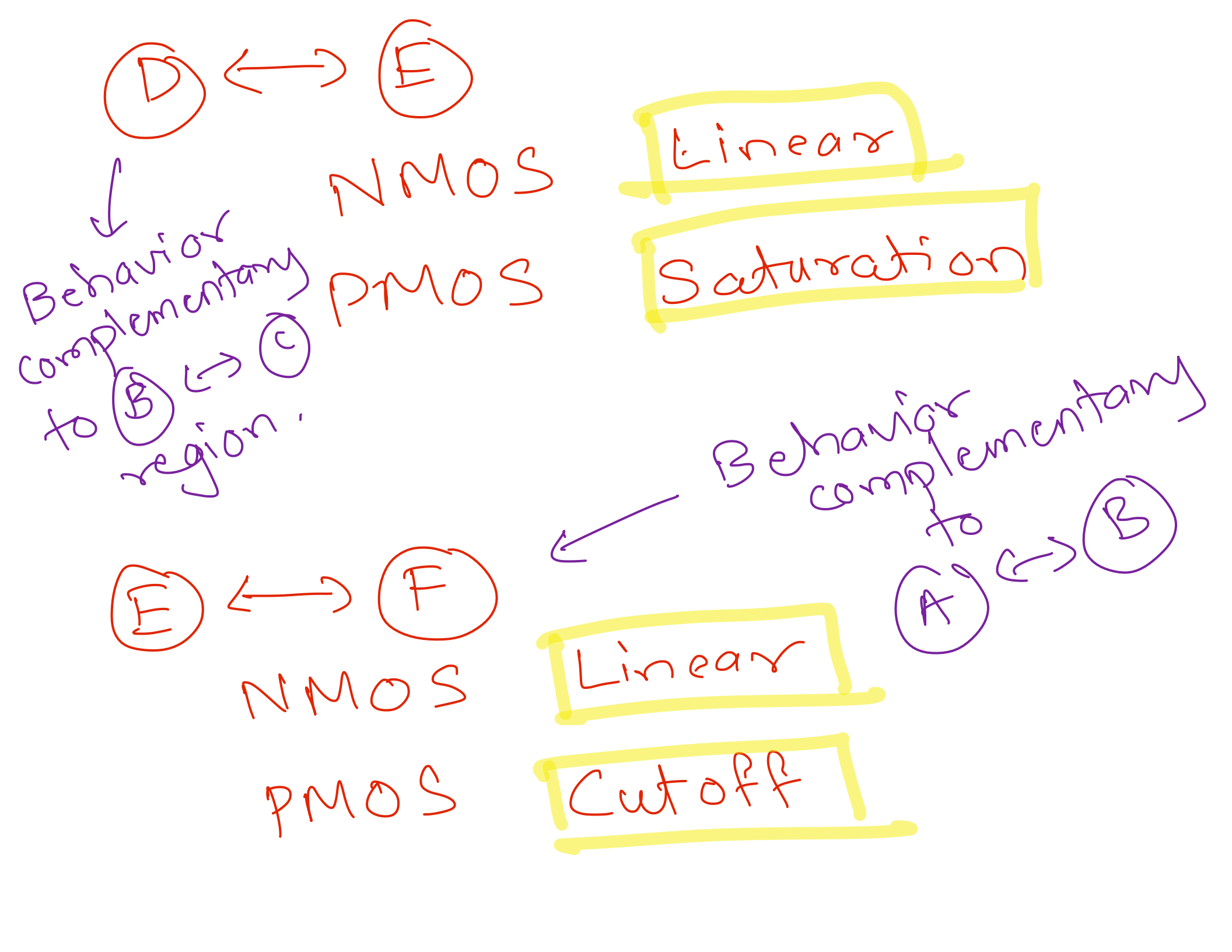
$V_{in} \approx V_{out}$

$|V_{GS}| \approx |V_{DS}|$

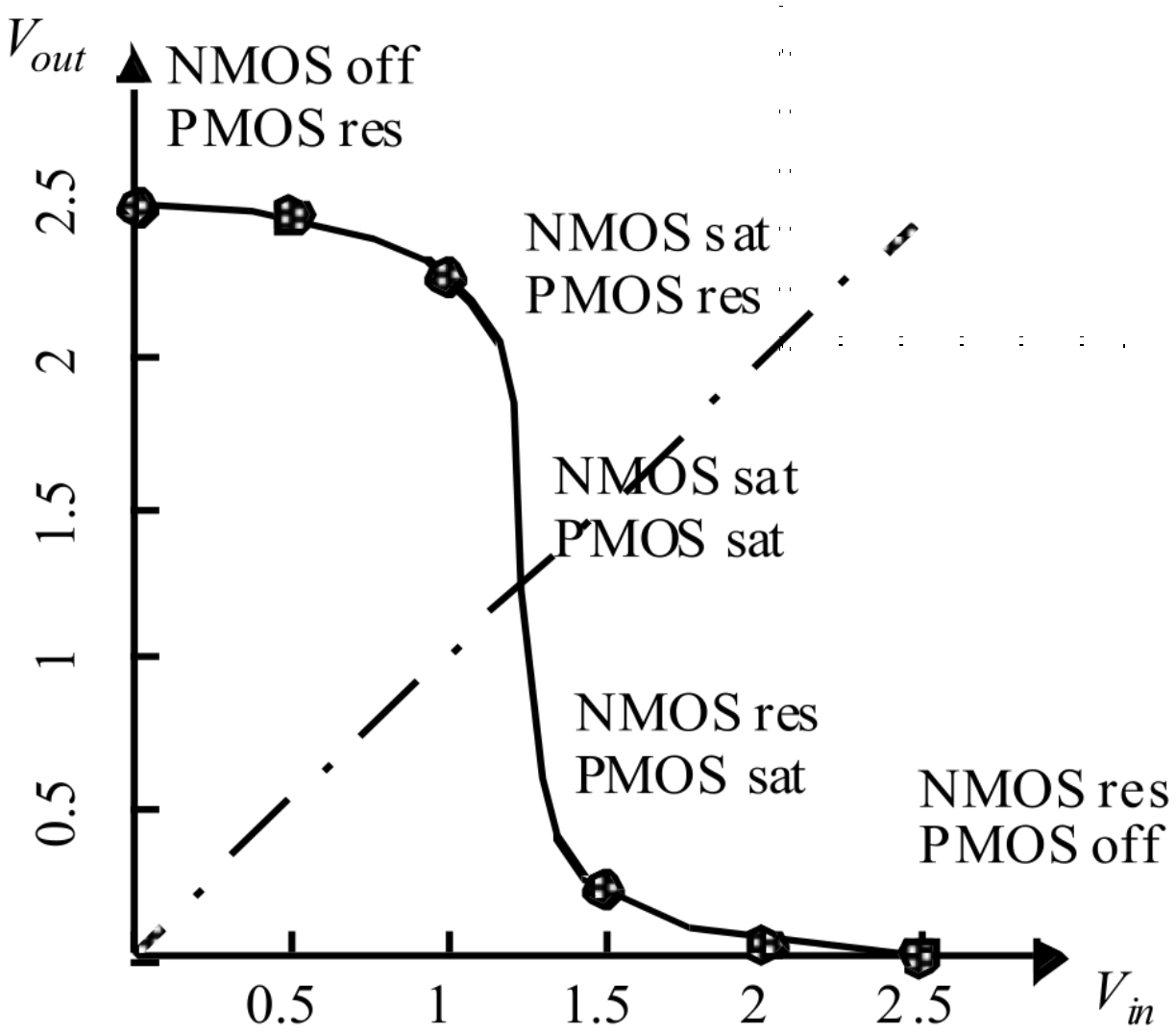
} At the intersection of VTC and 45° line

NMOS Saturation

PMOS Saturation.


D
E
NMOS
Linear
Saturation
Behavior complementary to B
C
region.
PMOS
Behavior complementary to A
B
E
F
NMOS
Linear
PMOS
Cutoff

CMOS Inverter VTC



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Switching threshold as a function of transistor ratio

$$I_{Dn}(V_M) = I_{Dp}(V_M)$$

At the switching point

$$V_M = V_{in} = V_{out}$$

Both NMOS & PMOS are in saturation.

We use the V_{TH}^* model.

$$I_D = K' \frac{W}{L} \left(V_{GS} - V_{TH}^*\right) V_{DSAT}$$

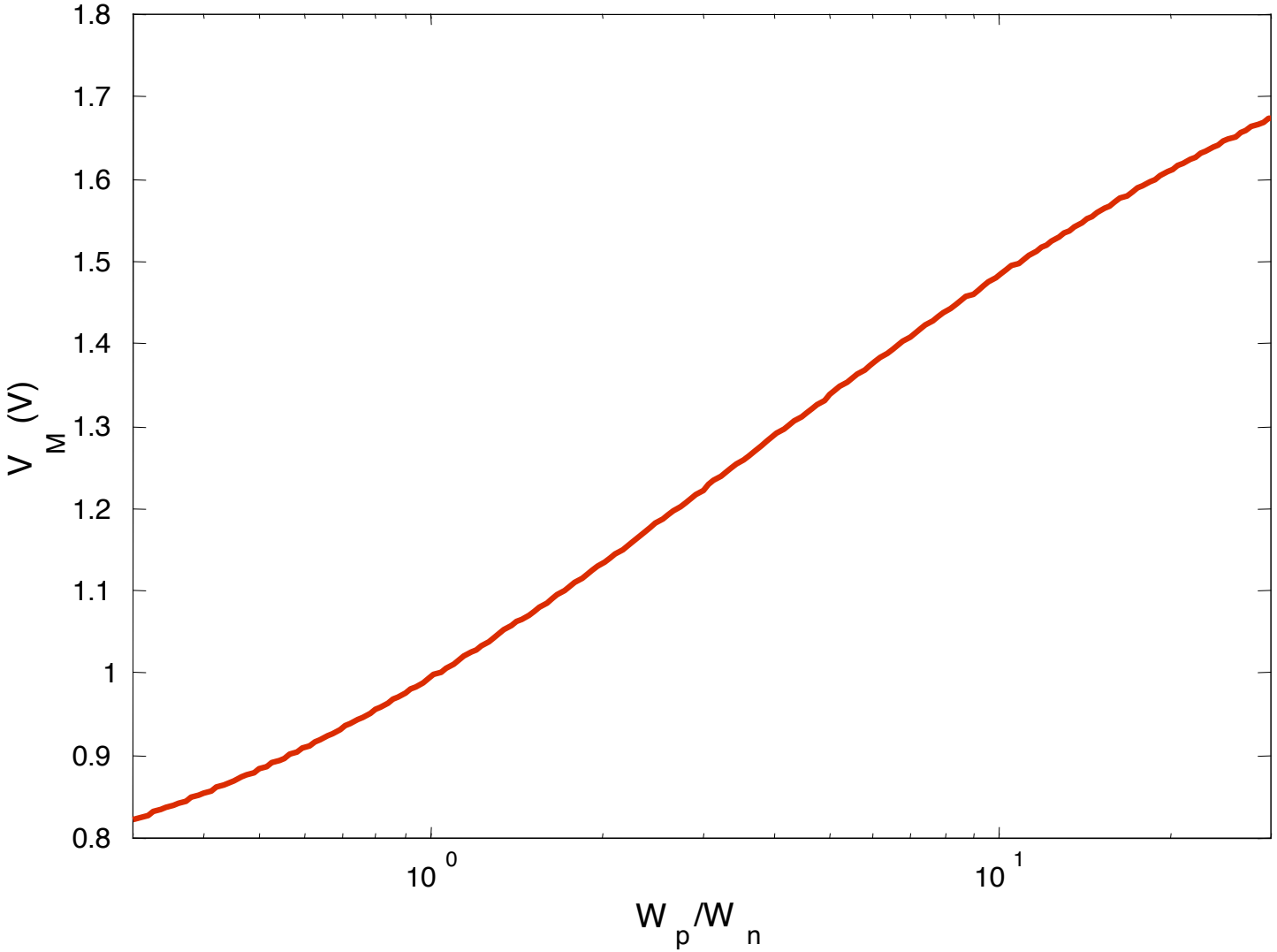
$$V_{DS_{v_{sat}}} = \frac{\mathcal{E}_c L}{2} \qquad \mathcal{E}_c = \frac{2 v_{sat}}{\mu}$$

$$\mu_{nmos} > \mu_{pmos}$$

$$\mathcal{E}_{c\,nmos} < \mathcal{E}_{c\,pmos}$$

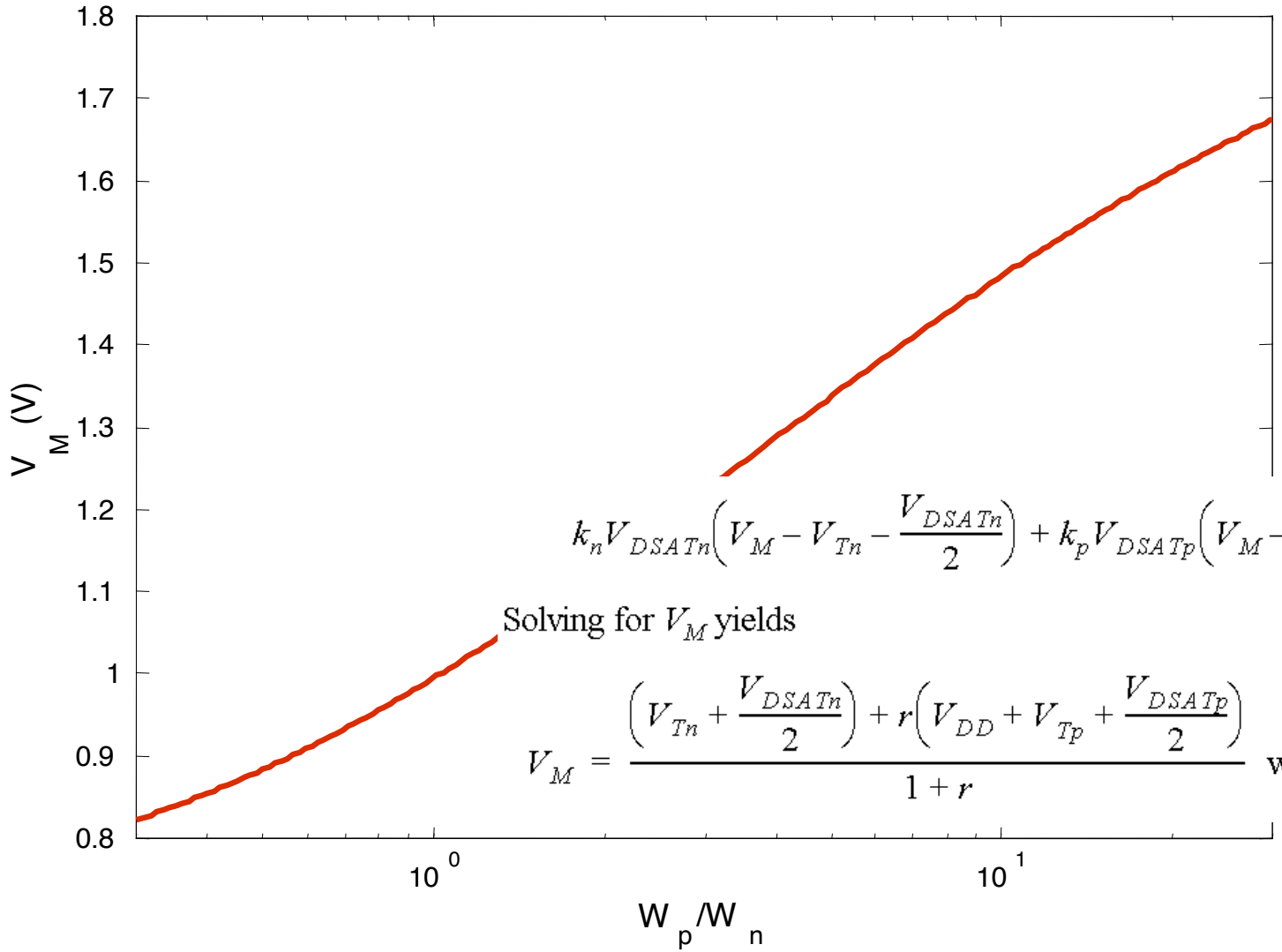
$$V_{DS_{v_{sat}\,nmos}} \neq V_{DS_{v_{sat}\,pmos}}$$

Switching Threshold as a function of Transistor Ratio



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Switching Threshold as a function of Transistor Ratio



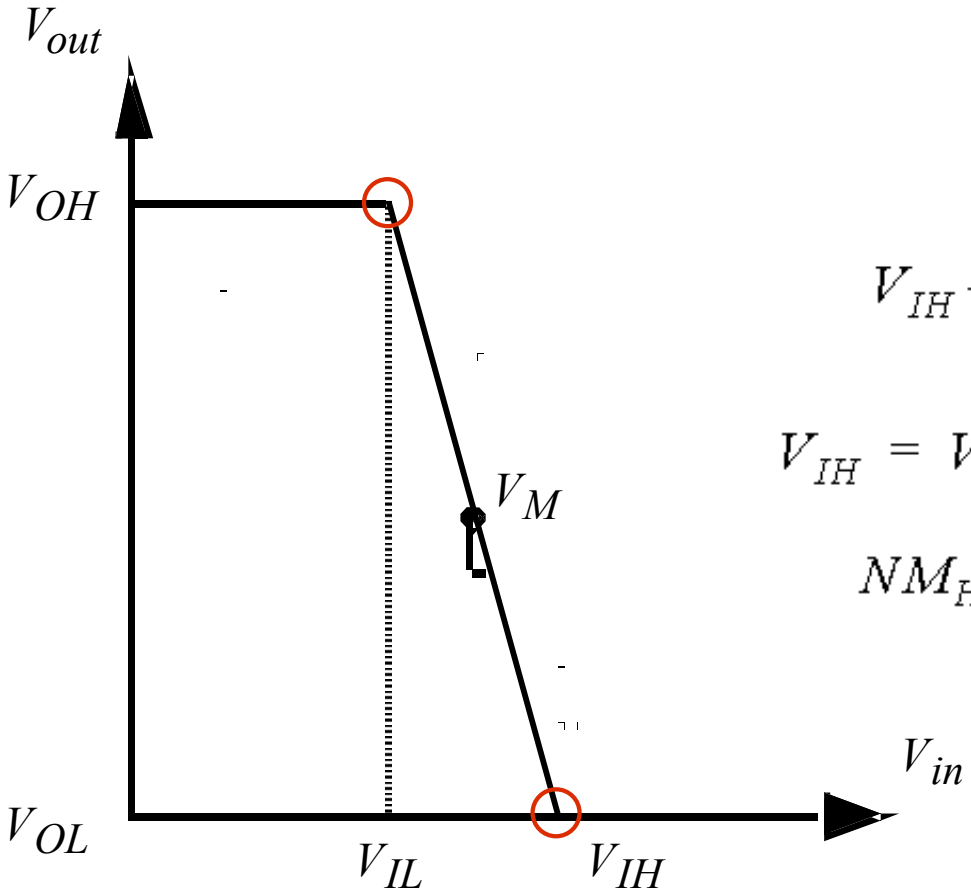
$$k_n V_{DSATn}\left(V_M - V_{Tn} - \frac{V_{DSATn}}{2}\right) + k_p V_{DSATp}\left(V_M - V_{DD} - V_{Tp} - \frac{V_{DSATp}}{2}\right) = 0$$

Solving for V_M yields

$$V_M = \frac{\left(V_{Tn} + \frac{V_{DSATn}}{2}\right) + r\left(V_{DD} + V_{Tp} + \frac{V_{DSATp}}{2}\right)}{1 + r} \quad \text{with} \quad r = \frac{k_p V_{DSATp}}{k_n V_{DSATn}} = \frac{\upsilon_{satp} W_p}{\upsilon_{satn} W_n}$$

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Determining V_{IH} and V_{IL}



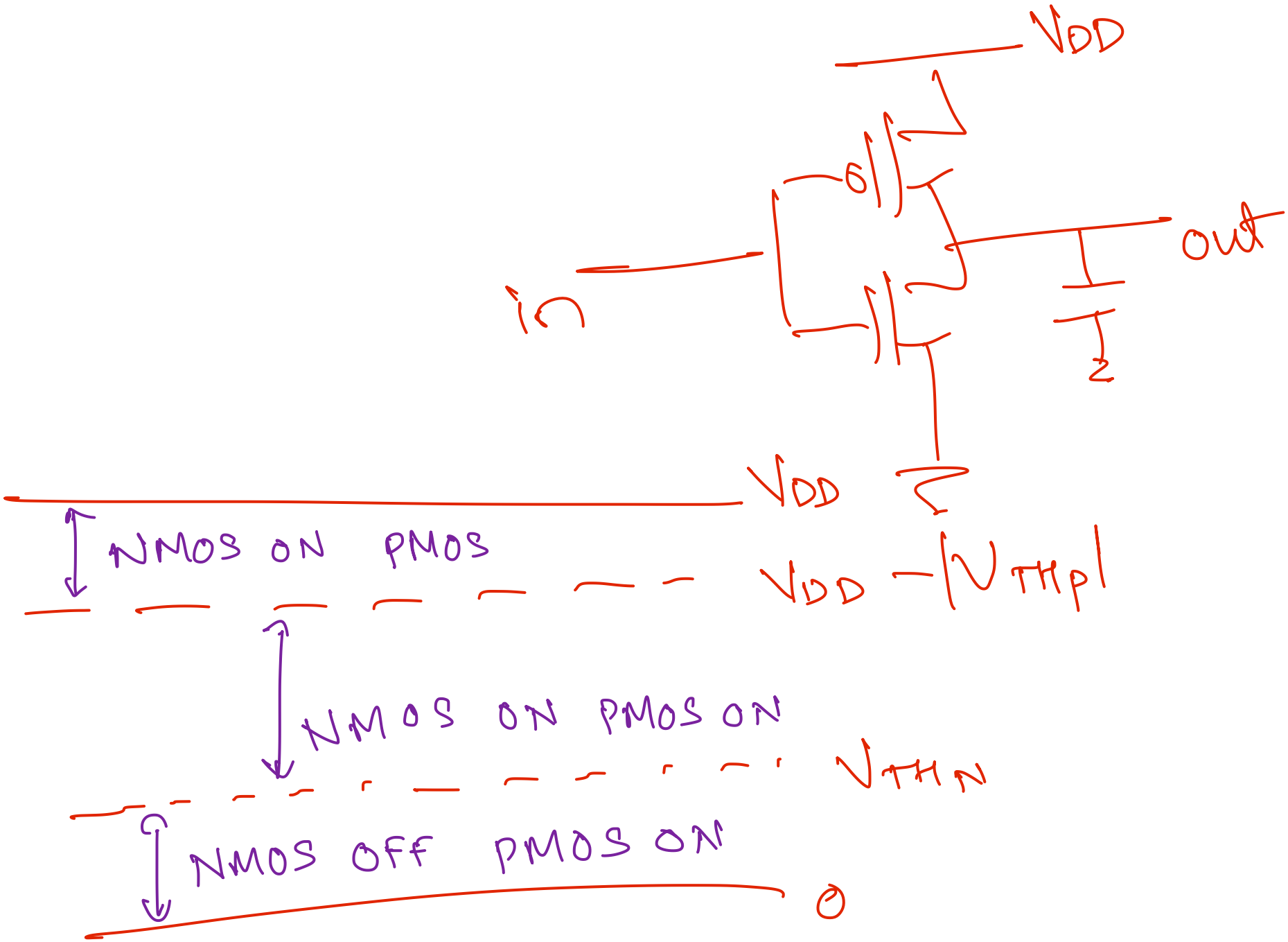
$$V_{IH} - V_{IL} = -\frac{(V_{OH} - V_{OL})}{g} = \frac{-V_{DD}}{g}$$

$$V_{IH} = V_M - \frac{V_M}{g} \qquad V_{IL} = V_M + \frac{V_{DD} - V_M}{g}$$

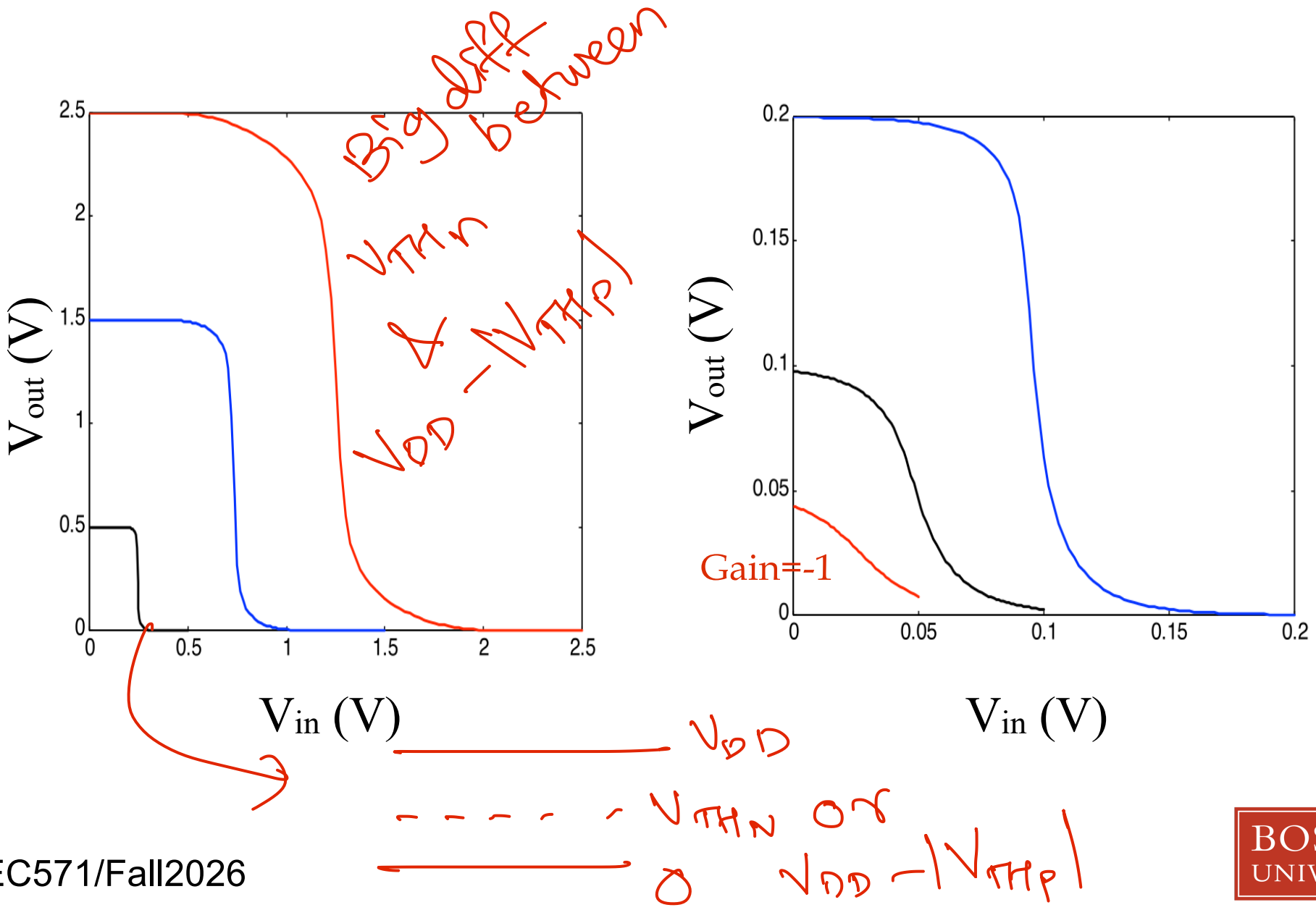
$$NM_H = V_{DD} - V_{IH} \qquad NM_L = V_{IL}$$

A simplified approach

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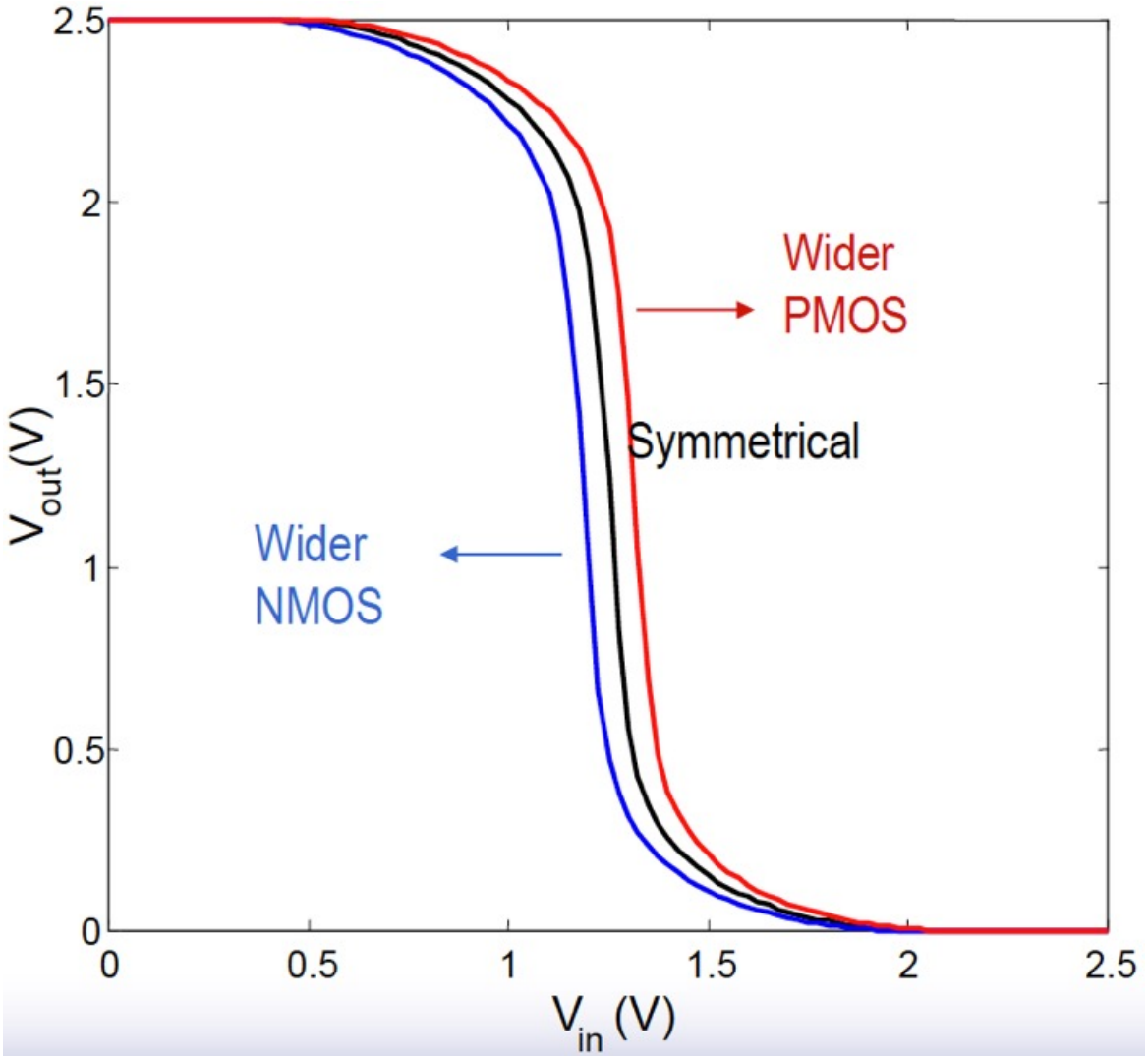

VDD
6
out
in
2
VDD
NMOS ON PMOS
VDD - |VTHp|
NMOS ON PMOS ON
VTHN
NMOS OFF PMOS ON
0

Gain as a function of V_{DD}



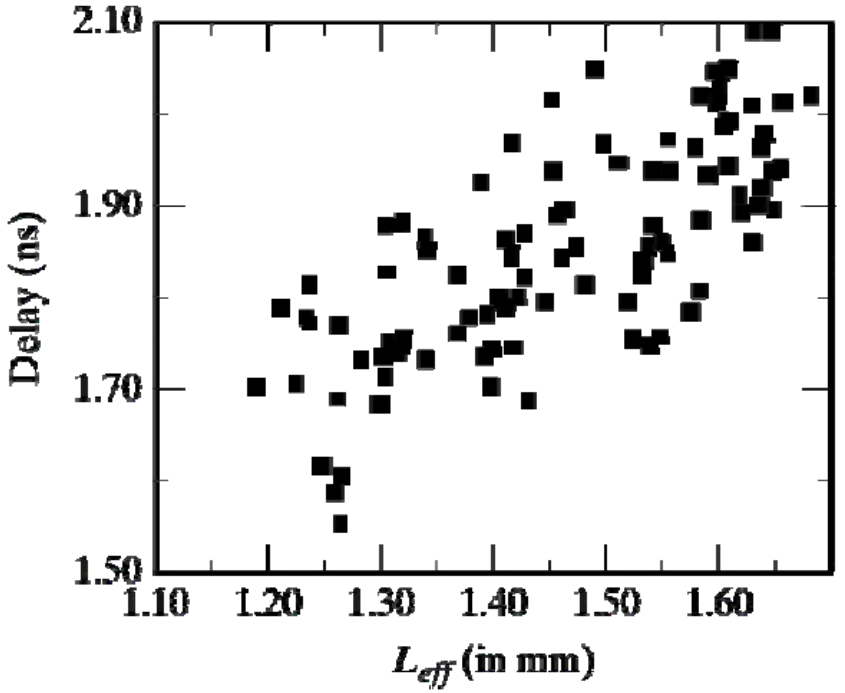
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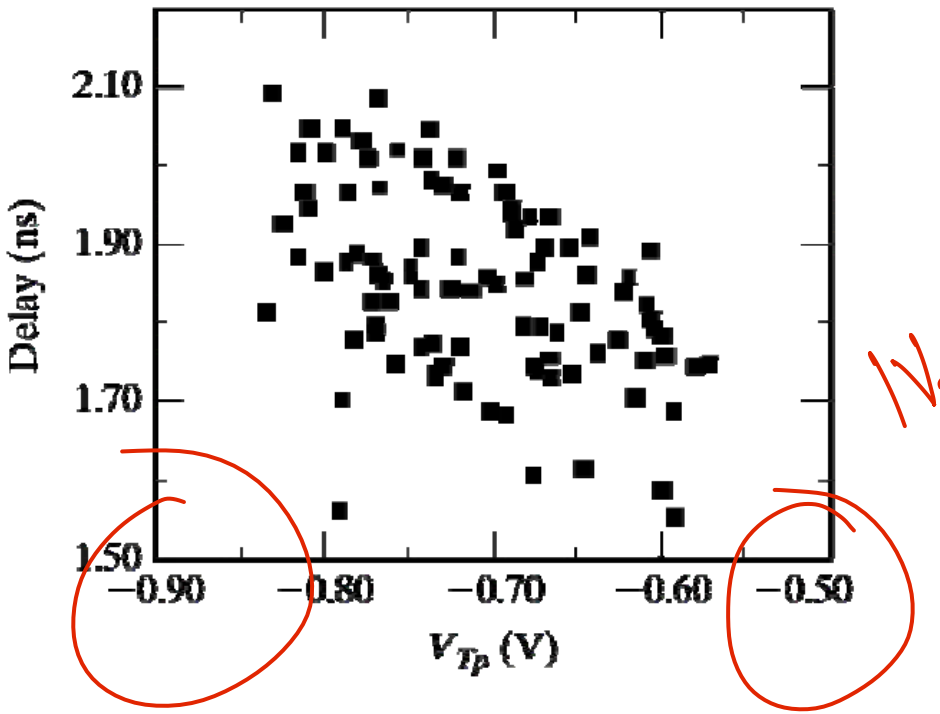
Impact of Sizing



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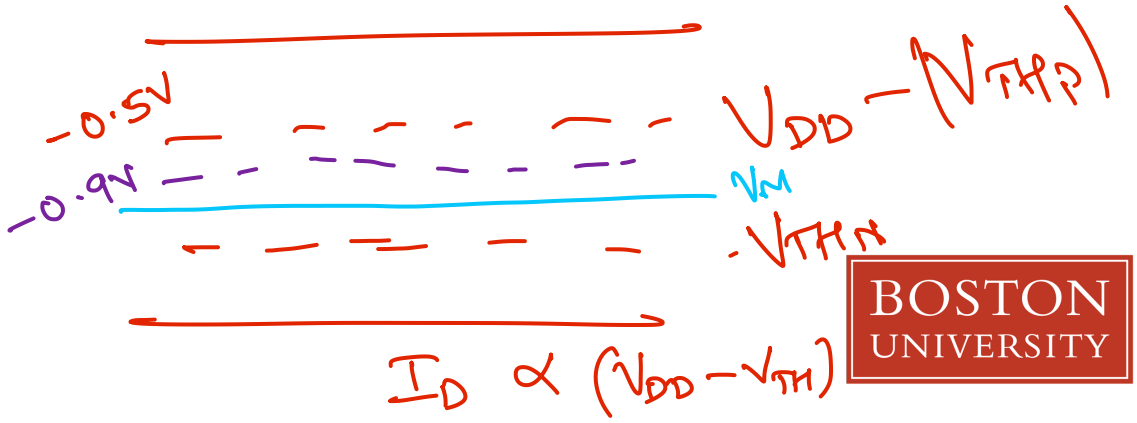
Process variations





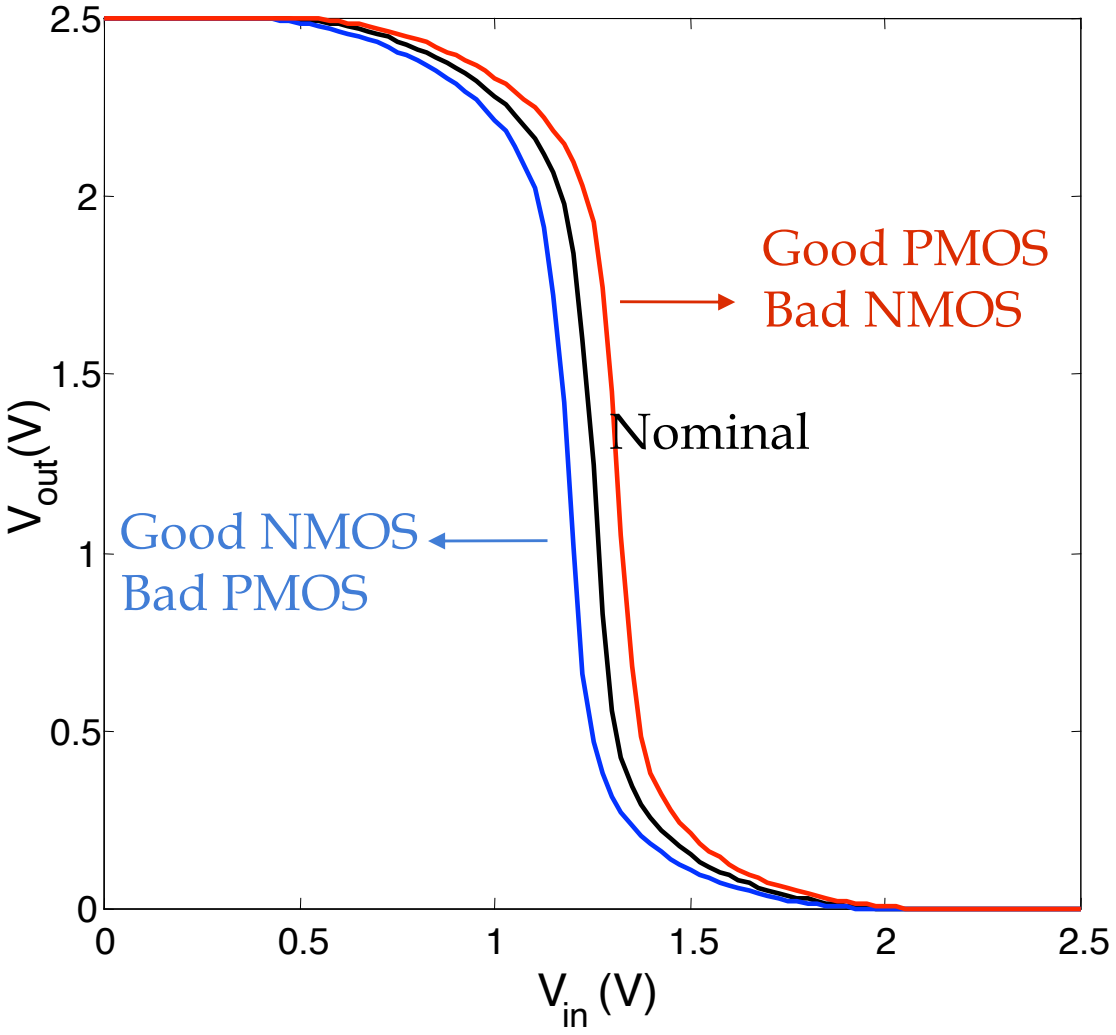
- Process corners - SS, ST, SF, TS, TT, TF, FS, FT, FF

$I_{D_{0.9V}} < I_{D_{0.5}}$

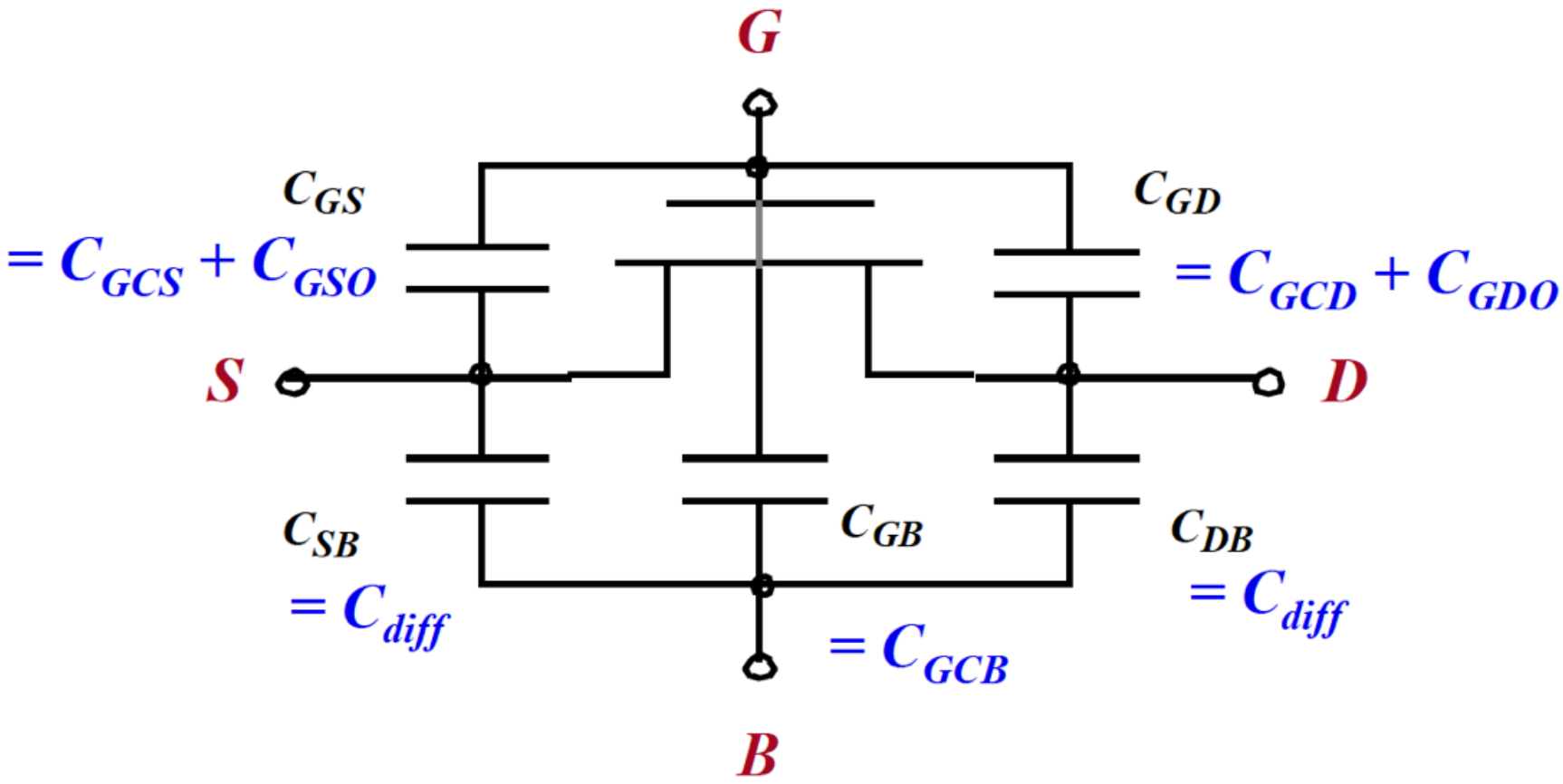


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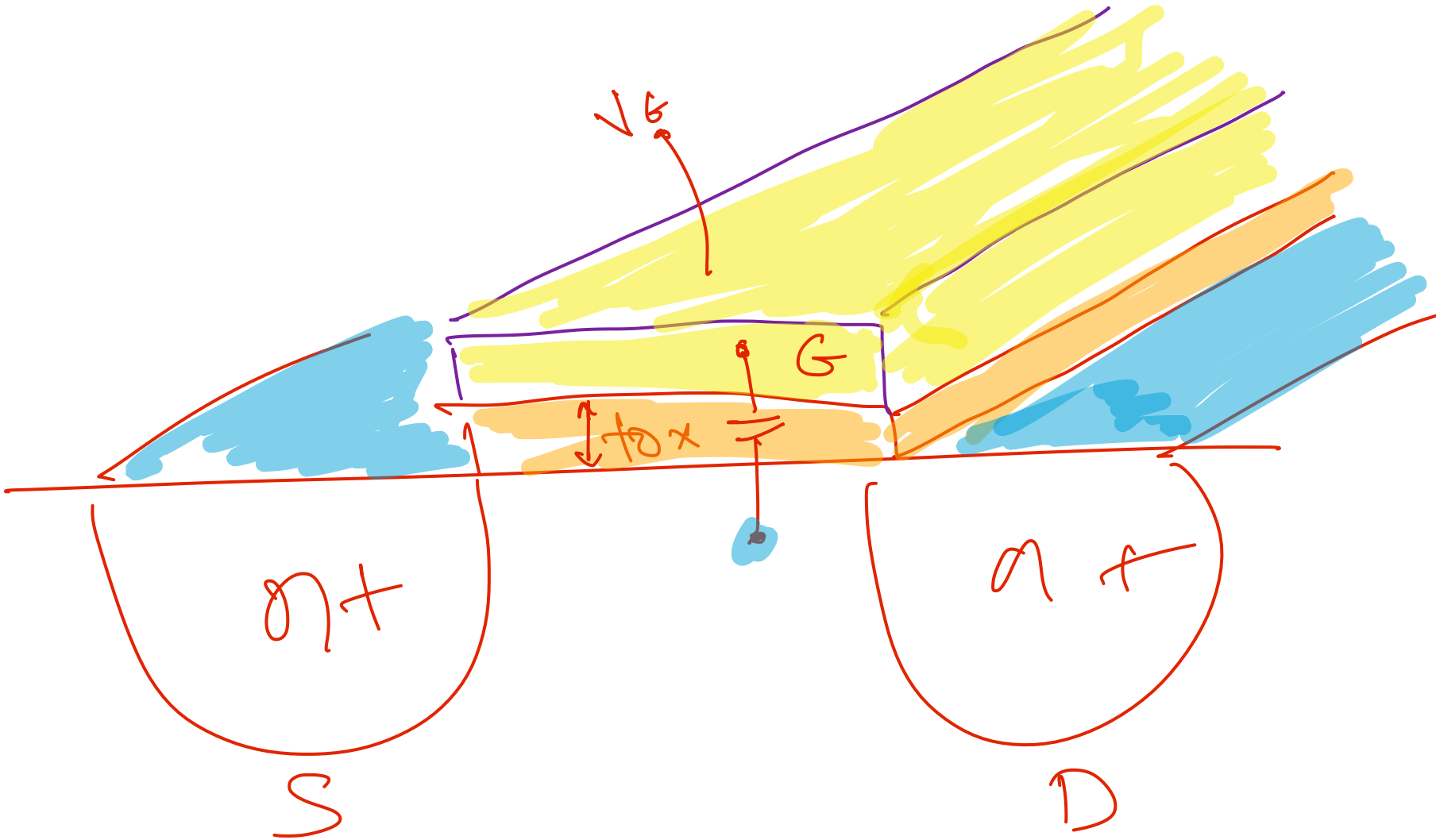
Impact of Process Variations



MOS Capacitances



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B

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} \quad (F/m^2)$$

Capacitance from gate to oxide

$= C_{ox} \cdot W \cdot L$

- Channel does not have a terminal
- Lump part of the cap to S, D & B terminals

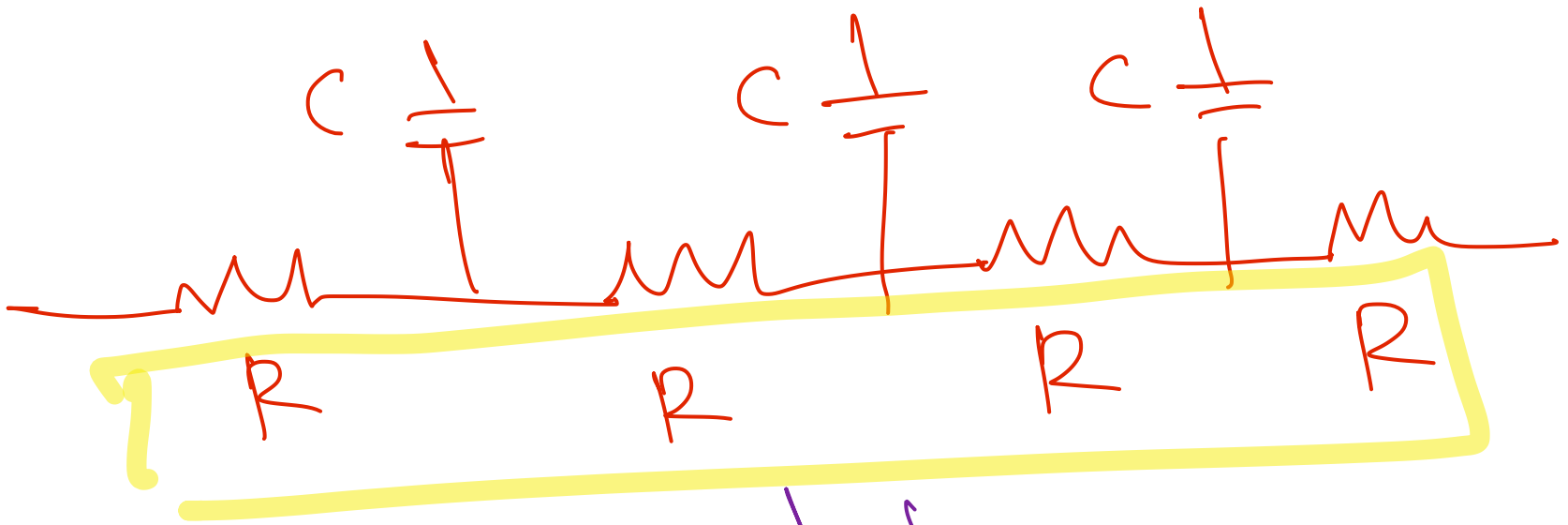
$$C_{GS} = C_{GCS} + C_{GSO}$$

C_{GCS}: channel capacitance component

C_{GSO}: overlap capacitance component

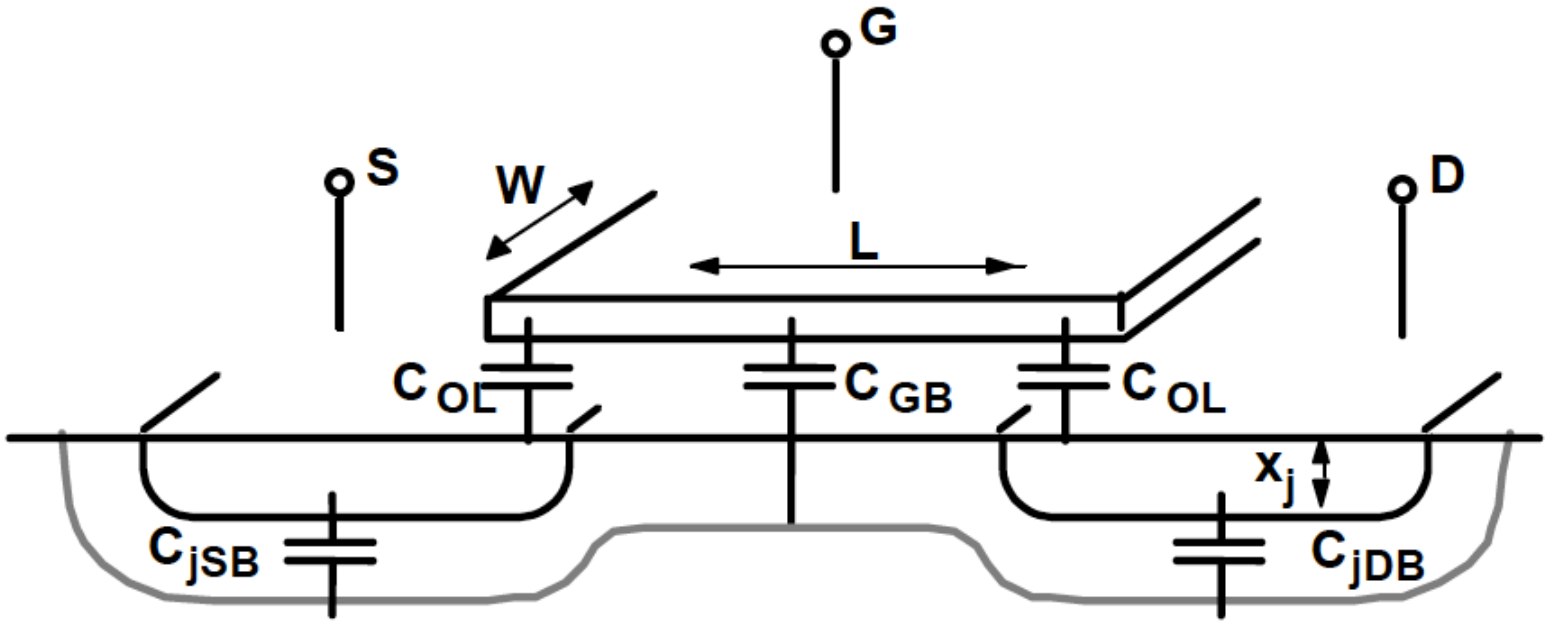
Model the xtor channel behavior using RC n/w.

Caps distributed across the channel



Channel charge.

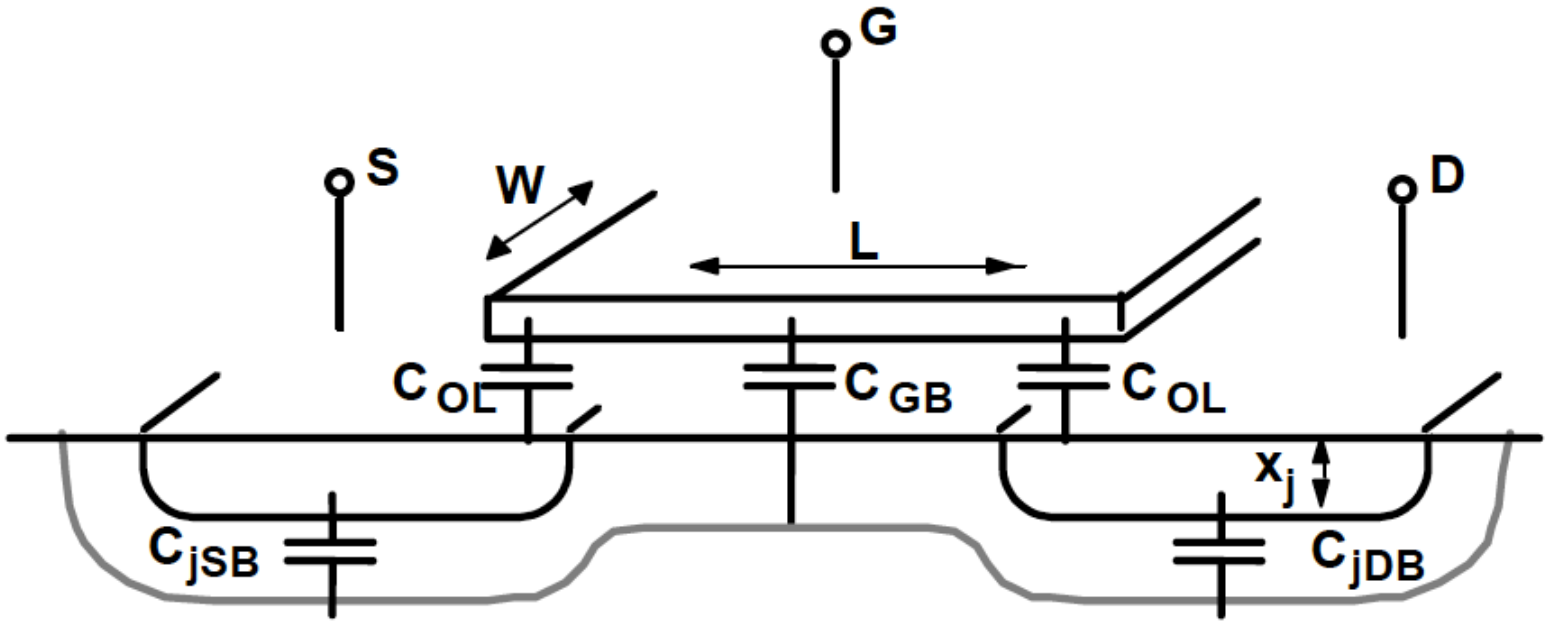
Transistor in cutoff



- No carriers in the channel to form the other terminal of the capacitor
  - Substrate acts as the other terminal
  - Capacitance becomes a series combination of gate oxide and depletion capacitance

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Transistor in cutoff (contd)



- $|V_{GS}| < |V_T|$, total C_{GCB} is smaller than $W.L.C_{ox}$ (approx $C_{GCB} = 0$)

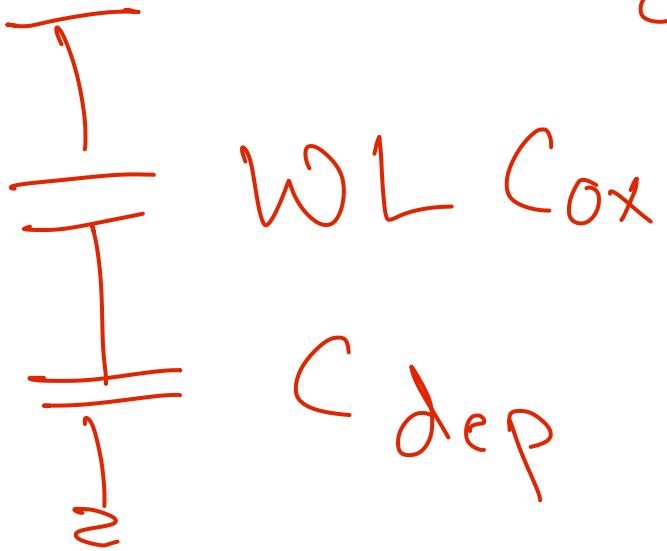
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Cutoff

$V_{GS} = 0$ $\quad C_{GB} = WLC_{ox}$

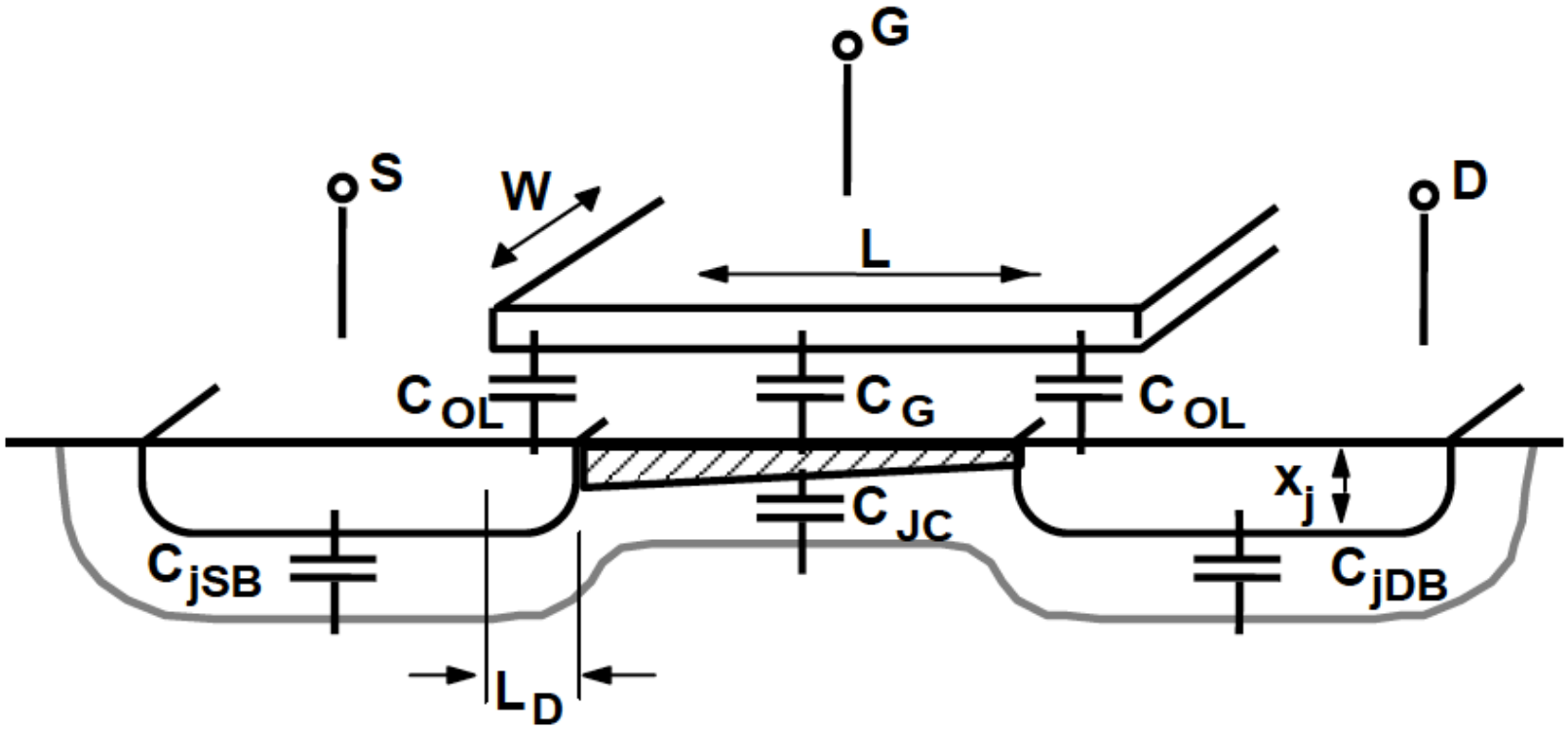
$V_{GS} \uparrow$ Depletion region forms under the gate.

$C_{dep} \uparrow$



Transistor in linear region

$V_{DS} \to 0$



- Channel is formed and acts as other terminal ($C_{GCB} = 0$)
- Split oxide cap equally between S & D
  - Changing either voltage changes channel charge

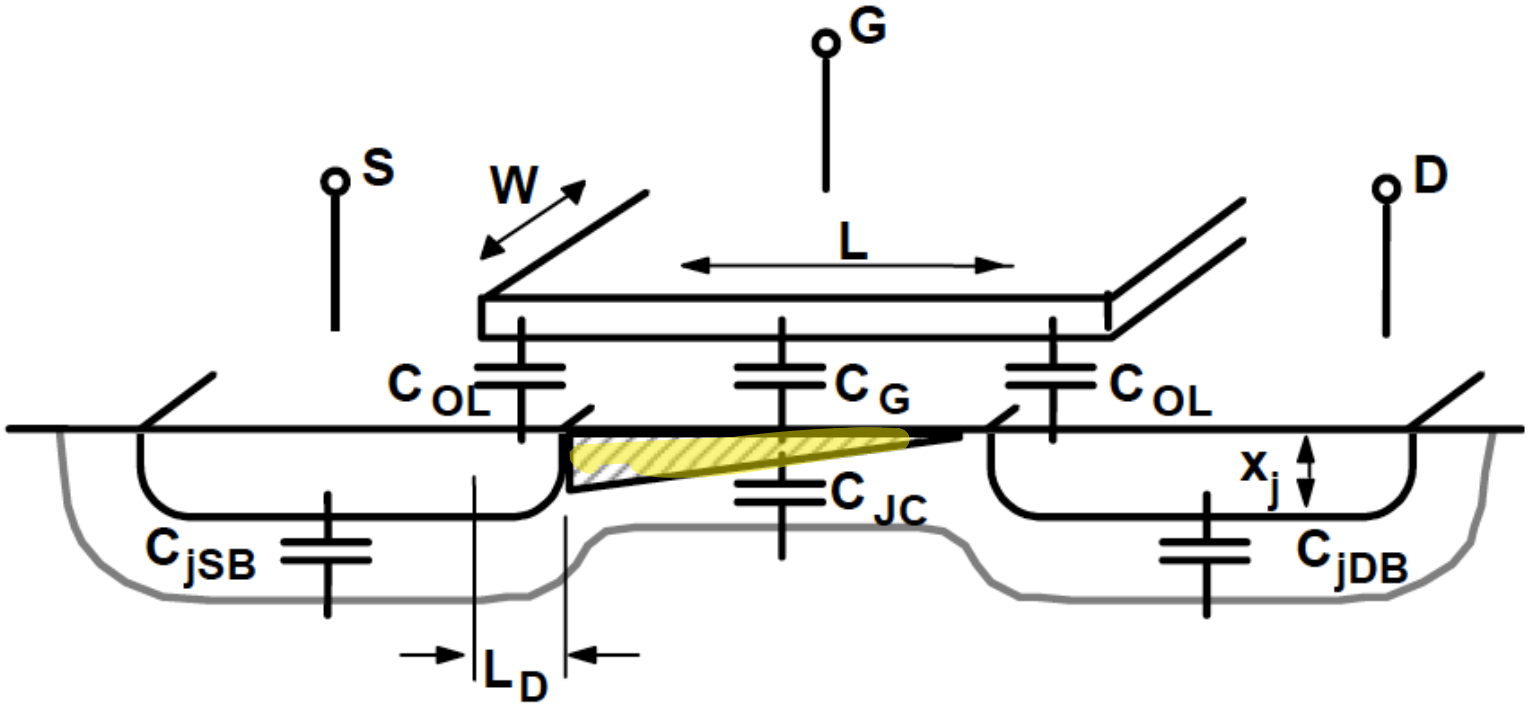
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Linear region



Resistance equally distributed across the channel.

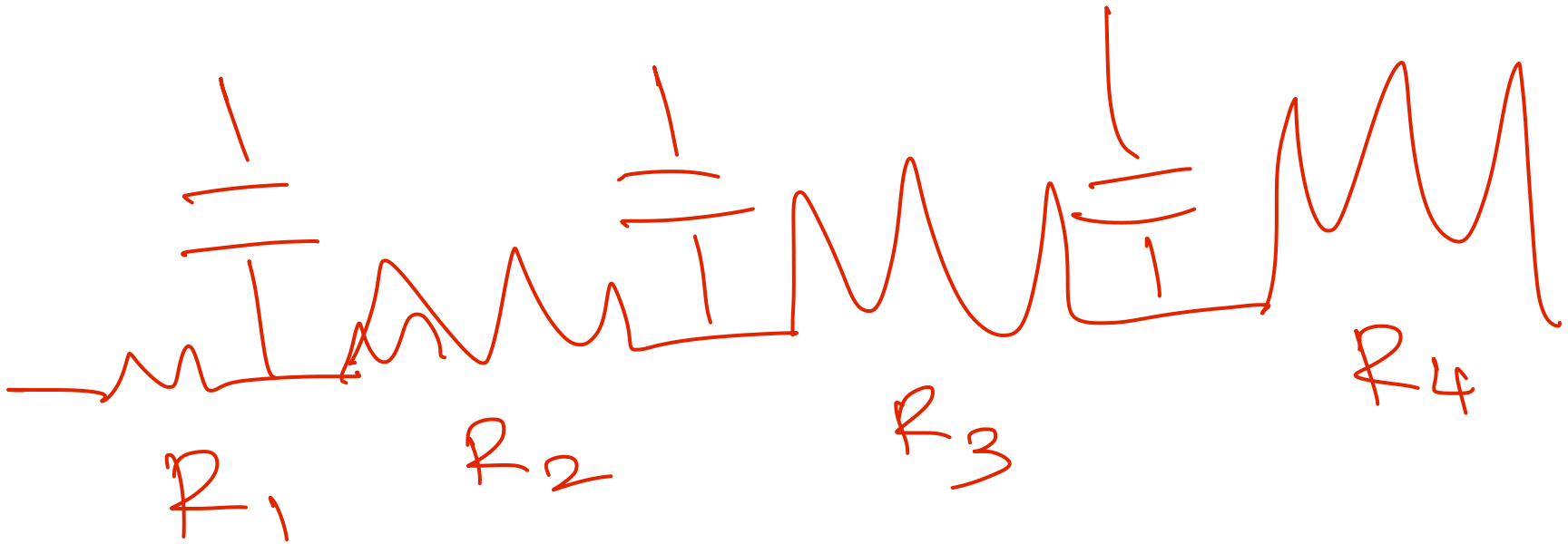
Transistor in saturation region



- $C_{GCS} \approx 2/3\ W.L.C_{ox}$
- Drain voltage no longer affects channel charge (set by source and V_{DS_sat})

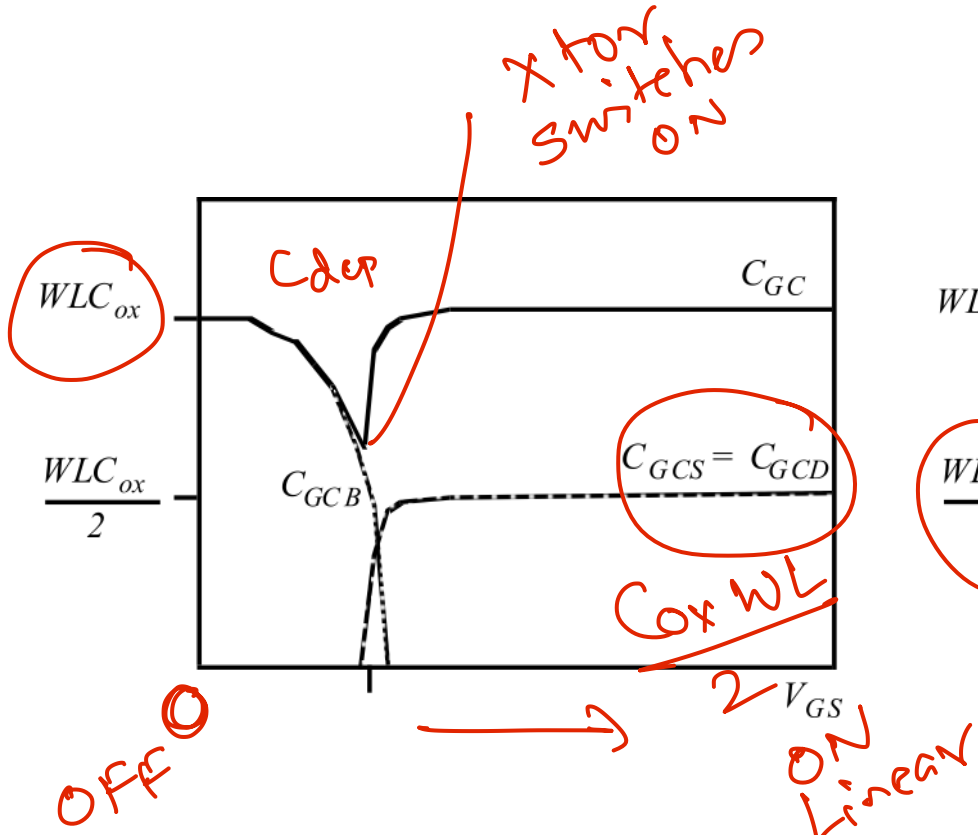
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Saturation

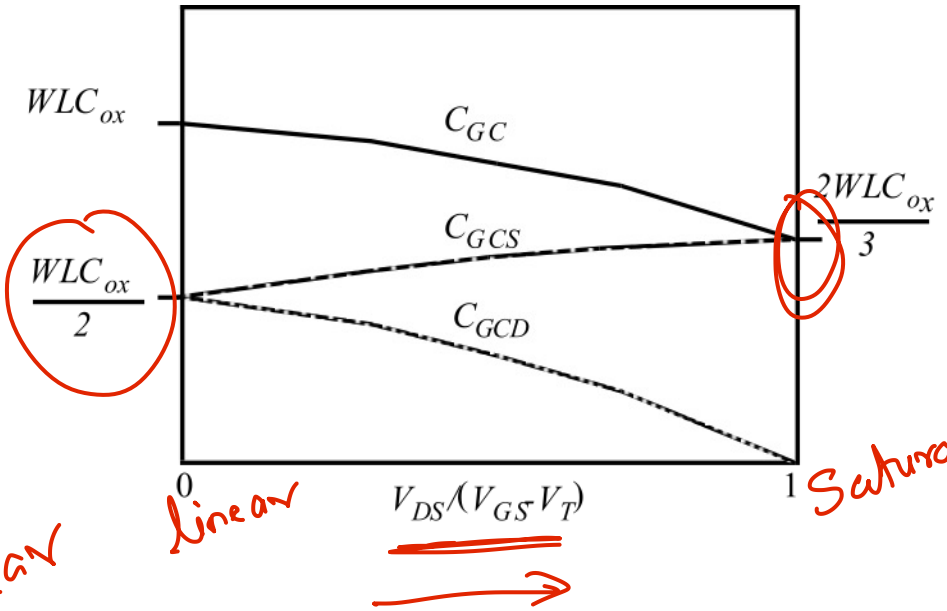


$R_4 >> R_1$

Gate capacitance



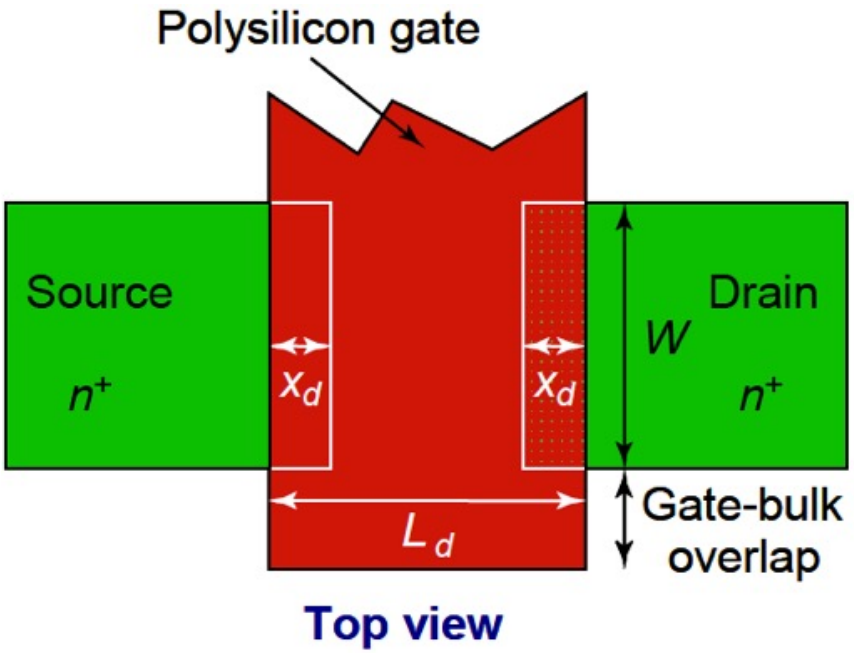
Capacitance as a function of V_{GS} (with $V_{DS} = 0$)



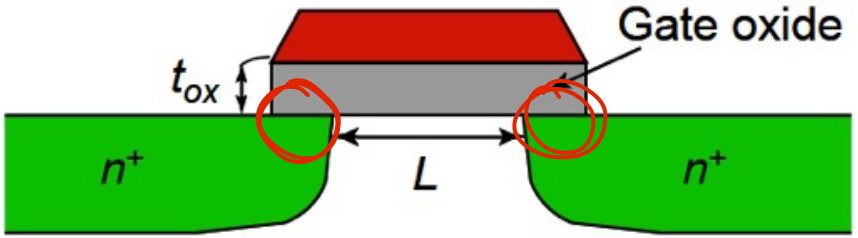
Capacitance as a function of the degree of saturation

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Gate overlap capacitance



Top view



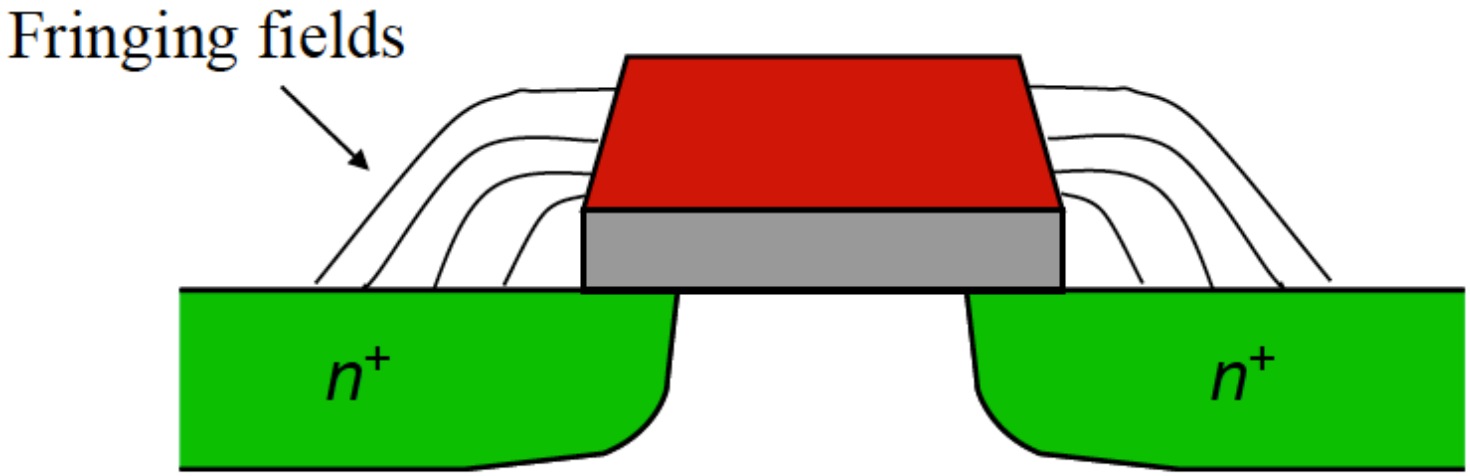
Cross section

$$C_O = C_{ox} \cdot x_d$$

Off/Lin/Sat → $C_{GSO} = C_{GDO} = C_O \cdot W$

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Gate Fringe Capacitance



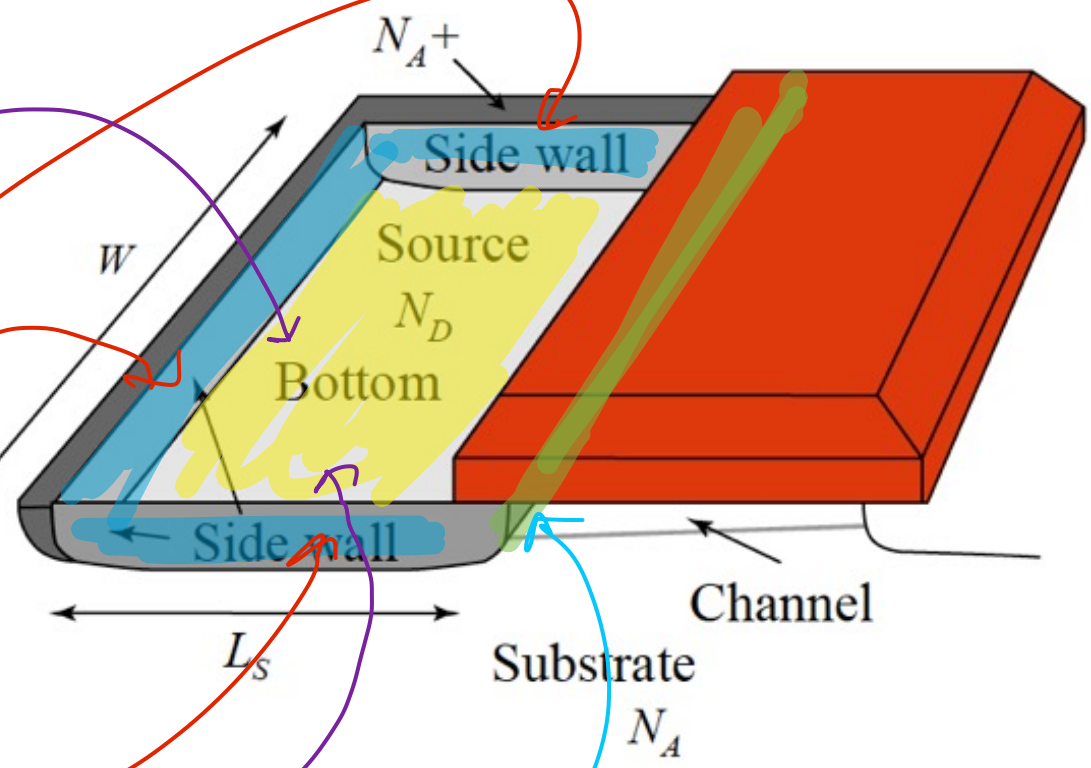
Cross section

- C_{ov} not just from metallurgical overlap - get fringing fields too
- Typical value - 0.2 fF/W

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Diffusion capacitance

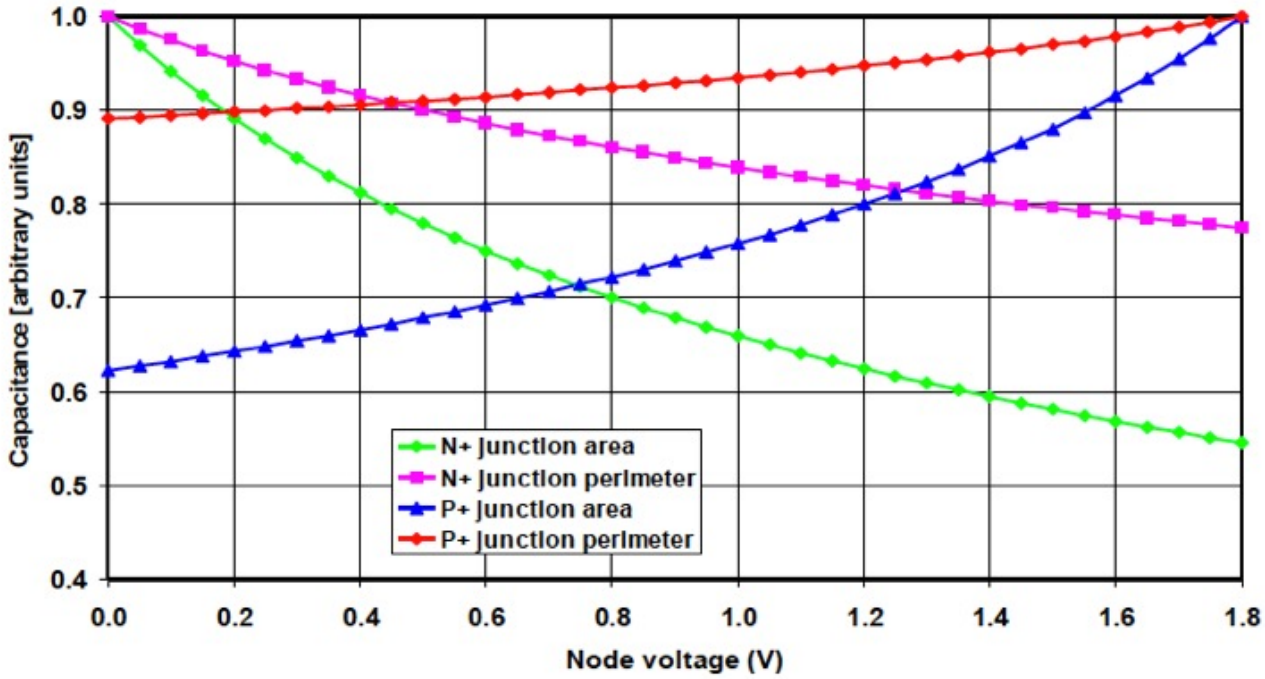
- Bottom
  - Area cap
  - $C_{bottom} = C_j \cdot L_S \cdot W$
- Sidewalls
  - Perimeter cap
  - $C_{sw} = C_{jsw} \cdot (2L_S + W)$
- GateEdge
  - $C_{ge} = C_{jgate} \cdot W$
  - Usually automatically included in the SPICE model



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Junction capacitance

- Junction caps are nonlinear
  - C_J is a function of junction bias



- SPICE model equations:
  - Area C_J = area × C_{J0} / (1+ $|V_{DB}|/\phi_B)^{mj}$
  - Perimeter C_J = perim × C_{JSW} / (1 + $|V_{DB}|/\phi_B)^{mjsw}$
  - Gate edge C_J = W × C_{Jgate} / (1 + $|V_{DB}|/\phi_B)^{mjswg}$

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Capacitance Model Summary

Gate-Channel Capacitance

- $C_{GC} \approx 0$ ($|V_{GS}| < |V_T|$)
- $C_{GC} = C_{ox} \cdot W \cdot L_{eff}$ (Linear)
  - 50% G to S, 50% G to D
- $C_{GC} = (2/3) \cdot C_{ox} \cdot W \cdot L_{eff}$ (Saturation)
  - 100% G to S

Gate Overlap Capacitance

- $C_{GSO} = C_{GDO} = C_O \cdot W$ (Always)

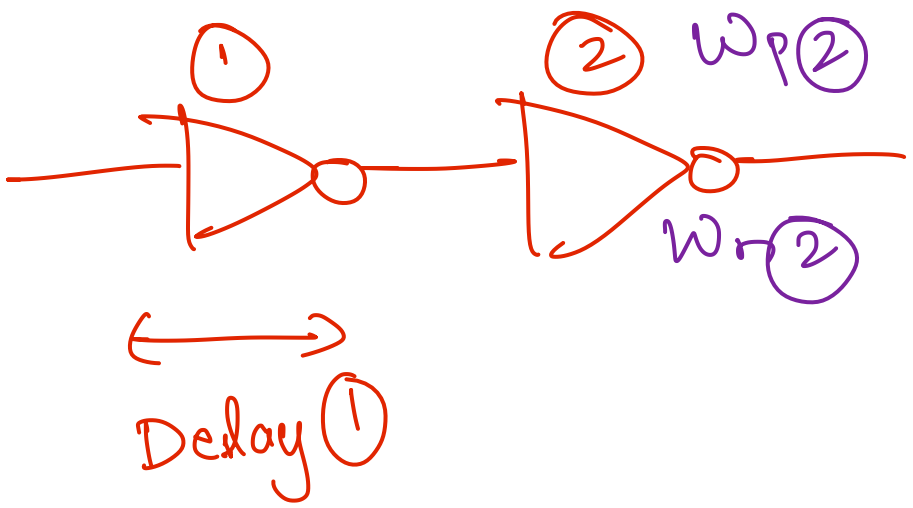
Junction/Diffusion Capacitance

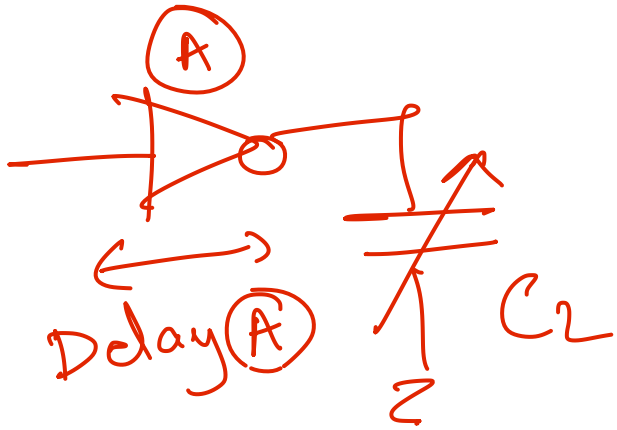
- $C_{diff} = C_j \cdot L_S \cdot W + C_{jsw} \cdot (2L_S + W) + C_{jg} W$ (Always)

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Alternate approach for calc.

Xtor cap





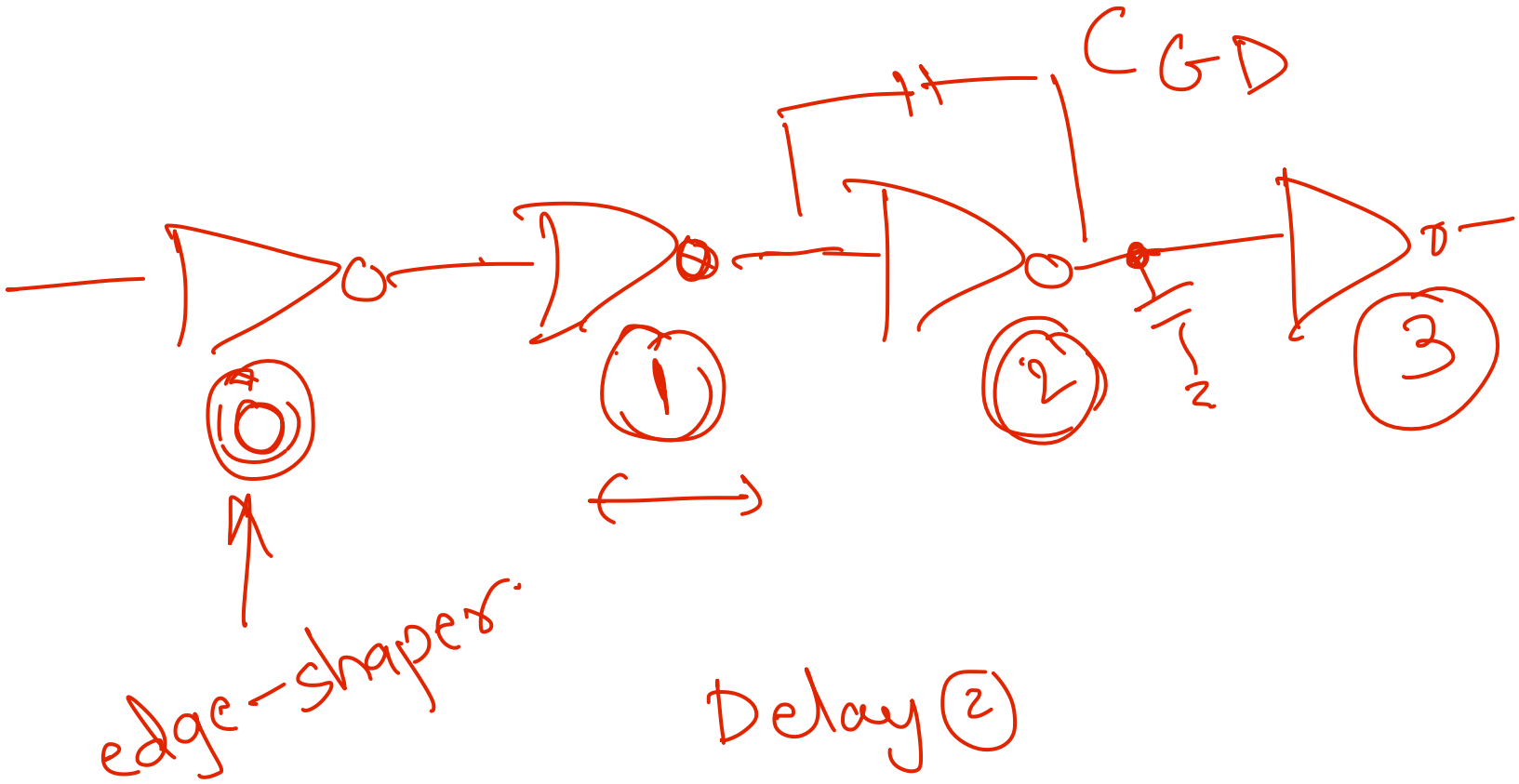
$C_{in_2} = C_2$ (tuned)

if Delay 1 = Delay A

$C_{in} (W_{p2} + W_{n2}) = C_{in_2}$

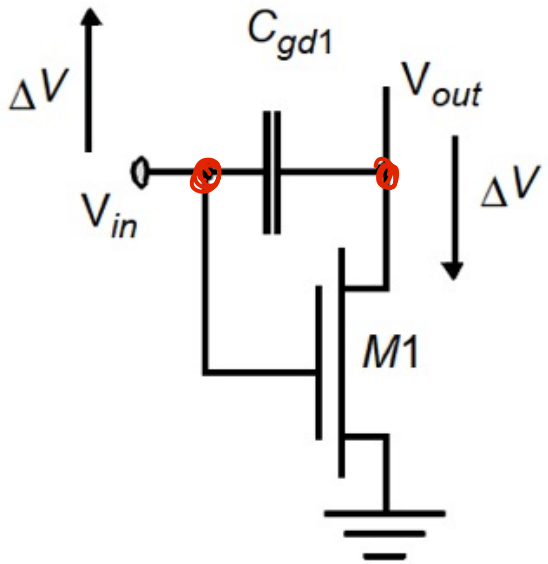
↑ known.

C_{in} → fF/μ

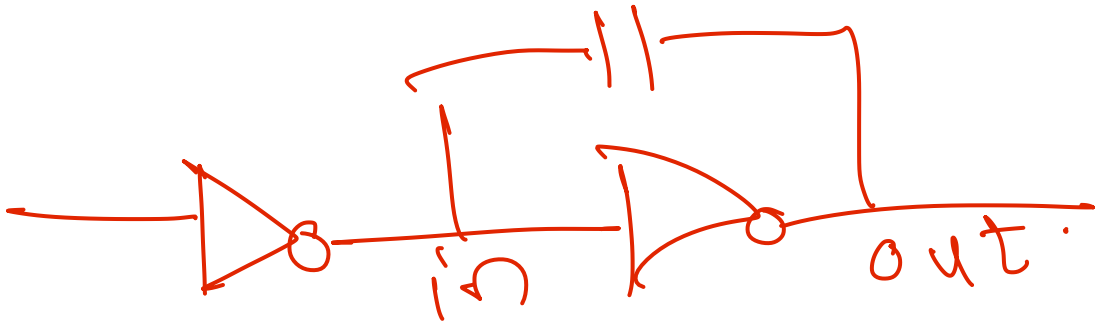

CGD
0
1
2
2
3
edge-shaper
Delay (2)

Miller effect

- As V_{in} increases, V_{out} drops
  - Once get into the transition region, gain from V_{in} to V_{out} > 1
- So, C_{gd} experiences voltage swing larger than V_{in}
  - Which means you need to provide more charge
  - Makes C_{gd} look larger than it really is
- Known as the "Miller Effect" in the analog world



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in $\Rightarrow$ 0 $\rightarrow V_{DD}$

out $\Rightarrow V_{DD} \rightarrow 0$

Effective change in voltage is $2V_{DD}$

$C_{eff} = 2\,C_{GD}$

Miller capacitance